WHITEPAPER · V2.6.3

Fabius

A plugin, not a platform — fifteen coordinated skills and twenty-two proven routing rules for supported agent harnesses, and the mathematics of knowing when to stop.

Ariel Shemesh

fabius-landing.vercel.app · public · proprietary · provenance-verifiable

ABSTRACT

Large language models, used as coding and reasoning agents, fail in two opposite directions: they over-explain and over-engineer, or — when told merely to "be concise" — they cut the validation, security, and accessibility that make an artifact correct. **Fabius** is a single operating stance, shipped as a plugin — fifteen coordinated skills under one router — that resolves this tension along one axis: *scout wide in what you investigate, strike narrow in what you ship.*

This whitepaper presents fabius as a system. It gives the architecture (a router — the *praetorium* — that dispatches on three axes over an always-on lean core, four engineering specialists, and nine domain verticals), the decision policy (a proven core of **twenty-two** routing rules — eighteen forming the mathematical core and four governing what the router itself adds: model-tier dispatch, long-horizon loops, vertical composition, and corpus externalization — plus a researched frontier layer at the working edge), and — the centre of the document — the **mathematics** beneath each core rule: every gate reduced to a formal statement, proved, and adversarially checked. We then prove the core *coheres*: all twenty-two rules are one expected-loss / value-of-information threshold applied to disjoint variables, composed in an acyclic pipeline that routes every task and terminates — every gate adversarially checked, the composition's exceptions printed in full. A strict boundary separates fabius's proven mathematical gates from the few operational heuristics it states as such — and marks the researched frontier layer (R14–R16 · M10–M13) as the working edge, outside the proof by construction. Every rule here is fabius's own. We close with the **fabius benchmark** — one dated test read out on four panels: blind model-judged quality on the four Claude tiers used in the 2026-07-01 run; a mixed track whose historical code scores were reported by a tool-using model operator and whose factual checklists were interpreted by model graders; aggregate-only cross-family demos; and the **FBS run**, scored by model judges and a model factual-check grader. Panel A shortened all four measured tiers and improved three; the other panels retain both gains and regressions. Historical receipts omit candidate answers, Panel B execution artifacts, and Panel C raw data, so they support aggregate replay, not independent re-judgment.

Fabius ships as a **plugin** — one set of operating rules for supported agent harnesses and model families. The model supplies capability; fabius supplies discipline through the loop *Scout* → *Plan* → *Strike* → *Prove* → *Record*. An optional repo-local runtime supplies orchestration checks when run from a full checkout; it is not a standalone package. The contract in one line: the user chooses the goal; fabius chooses the machinery — work is named capability-first and filled from whatever the harness exposes, research continues only while the expected value of the next action clears its cost, and stops on decision stability. The repository is public and proprietary, with recomputable provenance controls. This paper formalizes the decision policy — the router — at the plugin's core.

Version 2.6.3 · August 2026 · figures computed (numpy → SVG), reproducible

1 · Introduction

A capable model is not the same as a disciplined one. Hand a frontier LLM a real engineering task and it tends to do too much: it pads the explanation, builds a plugin system for a single flag, hedges every claim. Correct the over-building with the obvious instruction — "*be concise, write minimal code*" — and a second failure appears: the model now cuts the input validation, the error handling, the accessibility, the security. Brevity is not discipline. It trades one defect for another.

Fabius is named for Quintus Fabius Maximus, whose doctrine against Hannibal was to **scout the whole field and fight only the battle that mattered** — never the pitched engagement the stronger enemy wanted. Translated to an agent, the doctrine is a single axis that dissolves the thoroughness-versus-minimalism tension:

Scout wide in what you investigate. Strike narrow in what you ship.

Fan out to understand and to verify; deliver the smallest correct artifact; explain it in the fewest exact words. These never conflict, because they live on different axes — process and memory make you wide, lean makes you narrow. Fabius realizes this as one plugin of fifteen coordinated skills: a router, an always-on lean core, four engineering specialists (process, design, agent-engineering, memory), and nine domain verticals (go-to-market, defensive security, game craft, on-chain + sealing, automation, science, AI/ML engineering, markets & finance, cross-model council).

The contribution of this whitepaper is to show that the stance is not a vibe. Section 2 gives the system architecture and the single-owner coordination contract that keeps fifteen skills from contradicting one another. Section 3 states the decision policy: a proven core of twenty-two operational rules, each derived from fabius's own decision-theoretic core, and a researched frontier layer at the working edge. Section 4 — the heart of the document — supplies the **mathematics** under each core rule, every theorem proved and adversarially verified, with a hard line between the mathematical gates that *govern* a routing decision and the few operational heuristics stated as such. Section 5 proves the twenty-two core rules **compose into one coherent decision system**. Section 6 reports the dated fabius benchmark in four panels: blind model-judged quality, mixed execution/checklist verification, aggregate-only cross-family demos, and the model-graded FBS run on the versioned suite. The discipline of the whole is in Section 7: **measured, not claimed**.

2 · The system

Fabius is one plugin under one router, not a bundle of unrelated skills. A single router — the *praetorium* — dispatches fourteen coordinated capability layers over a thin supporting spine (fifteen skills in all, counting the router), so the agent gains end-to-end capability under one stance. The router does not just pick a layer; it dispatches on three axes at once: **which layer(s)**, **how much machinery** (the capability ladder of §4.1), and **which model tier**.

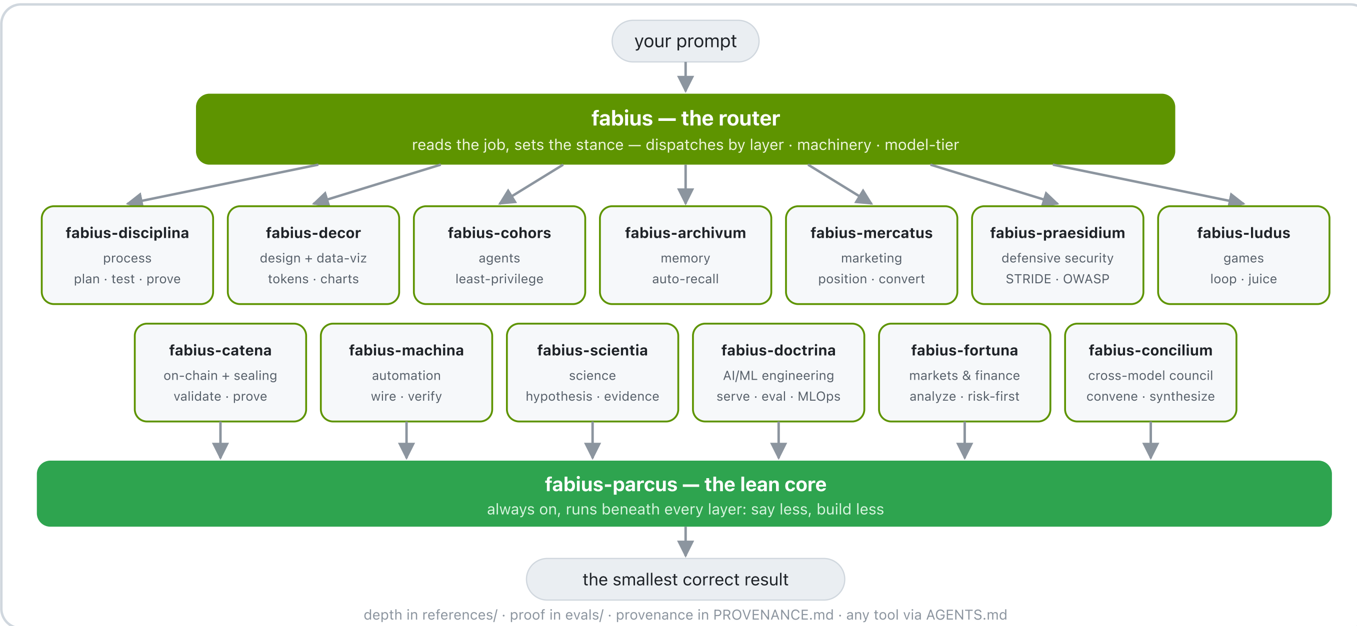

Figure 0 — the layer model. Your prompt enters the `fabius` router (the *praetorium*), which dispatches by layer, machinery, and model-tier to the specialist a task needs — four engineering specialists (`disciplina` process, `decor` design + data-viz, `cohors` agents, `archivum` memory) and nine domain verticals (`mercatus` go-to-market, `praesidium` defensive security, `ludus` games, `catena` on-chain + sealing, `machina` automation, `scientia` science, `doctrina` AI/ML engineering, `fortuna` markets + finance, `concilium` cross-model council) — all running on the always-on `fabius-parcus` lean core, producing the smallest correct result. Depth loads on demand from `references/` (indexed by `CORPUS.md`); the benchmark is in `evals/`; the stance ports to any tool via `AGENTS.md`.

2.1 · Fifteen skills, one stance

Each skill is a thin operating contract; its depth lives under `references/`, indexed by `CORPUS.md`, and loads only on demand.

Skill	Role	What it delivers
`fabius`	router · praetorium	reads the job, sets the stance, dispatches by layer + machinery + model-tier
`fabius-parcus`	lean core, always on	terse output · the YAGNI ladder · surgical changes · the never-trim floor
`fabius-disciplina`	engineering process	brainstorm → plan → test-first → prove · root-cause debugging
`fabius-decor`	design + data-viz	one-accent laws · token vocabulary · mobile-first · data-ink charts (figura) · decks + infographics + measured reports
`fabius-cohors`	agent engineering	definition schema · least privilege · five orchestration patterns
`fabius-archivum`	persistent memory	per-project memory · the LLM-wiki · index + log retrieval · recall · meeting transcripts into filed records
`fabius-mercatus`	go-to-market	positioning · message-to-awareness match · proof over adjectives · a one-action funnel · draft-only GTM outreach

Skill	Role	What it delivers
`fabius-praesidium`	defensive security	STRIDE per boundary · the OWASP pass · severity → fix → proof findings · AI-provenance-mark hygiene on content you own
`fabius-ludus`	game craft	the core loop first · deliberate juice · state as a machine · jam-sized scope
`fabius-catena`	on-chain + sealing	account-validation-first contracts (EVM + Solana) · money-safe transactions · verifiable provenance sealing
`fabius-machina`	automation	deterministic workflow glue · discover-from-live-schema → build → validate AND verify → activate
`fabius-scientia`	science	competing falsifiable hypotheses · source-grounded database lookups · reproducible field-standard pipelines
`fabius-doctrina`	AI/ML engineering	the model lifecycle — train/fine-tune → evaluate → serve → monitor · held-out eval + blind judges · vLLM-class serving · MLOps
`fabius-fortuna`	markets & finance	risk-first analysis · fundamental + technical + quantitative · valuation · honest out-of-sample backtesting (analysis, not advice)
`fabius-concilium`	cross-model council	convene N models on one question · first opinions → anonymized peer-review → chairman synthesis · ensemble epistemics, gated on cost (N+N+1 calls)

Latin, for the curious: **parcus**, frugal · **disciplina**, training · **decor**, what is fitting · **cohors**, the cohort · **archivum**, the record office · **mercatus**, the marketplace · **praesidium**, the garrison · **ludus**, the game and the school where you drill it · **catena**, the chain · **machina**, the working mechanism · **scientia**, knowledge won by method · **doctrina**, the body of learning (a machine taught to learn) · **fortuna**, fortune and the turn of the market · **concilium**, the summoned council.

2.2 · Single owner, zero overlap

The coordination contract is one rule: **each rule has exactly one owning layer; every other layer links to it instead of restating it.** Planning, test discipline, and the clarifying-question procedure are owned by `disciplina`; lean prose, the YAGNI ladder, and the never-trim guardrail list by `parcus`; routing and the kill-switch by `fabius`. The line between core and vertical is drawn the same way: `parcus` owns the *never-trim security floor* (don't cut validation), while `praesidium` owns the *active* security work (model the threat, name the check, prove it closed) and references that floor rather than restating it. `parcus` is the only always-on layer: it composes *underneath* whatever task layer the router selects and never competes for a task verb. This single-owner discipline is what makes the formal coherence of Section 5 possible — no two layers write the same rule with opposite intent.

2.3 · The corpus — the brain holds the index, not the library

The depth behind the skills — design references, an agent catalog with a vector index, a knowledge engine, and the marketing, security, and game-craft playbooks — is large. Fabius keeps the *discovered contract surface* lean by separating brain from library: a single fabius-branded index (`CORPUS.md`) sits over every capability library, and the router holds only the index. Recursive verification requires exactly fifteen top-level `SKILL.md` contracts; nested library documents use `REFERENCE.md`, so a host cannot accidentally discover them as extra skills. On a task the system reads the index and pages in the matching slice. The standing rule is that adding library depth is adding an indexed reference, not another discovered contract.

3 · The decision policy

A stance is only as good as the decisions it makes: *which* layer to route to, *when* to spend a tool call or a second agent, *how long* to keep refining, *what* to load from memory. Fabius's router carries an explicit policy of twenty-two proven rules — thirteen routing rules (R1–R13) and nine orchestration / memory rules (M1–M9) — each derived from fabius's own decision-theoretic core (R13 from its single-owner contract), plus a researched frontier layer (R14–R16 · M10–M13) documented in the routing policy.

3.1 · Route by classification, then climb one rung

Specialist selection is reproducible, not vibes. Fabius opens every non-trivial task by naming its load on three axes — **Memory, Tools/Action, Planning** — the universal capability axes fabius routes on **(R1)**. Memory routes to `archivum`, Tools to `cohors`, Planning to `disciplina`; zero axes loaded stays in the lean `parcus` core. A task in a named vertical also routes on a **Domain** axis to its owner — UI/data-viz to `decor`, go-to-market to `mercatus`, defensive security to `praesidium`, games to `ludus`, on-chain/provenance to `catena`, automation to `machina`, science to `scientia` — where *domain picks what* while the three load axes pick *how*, and the vertical runs as a studio (R13). The classification *is* the routing rationale.

Then it climbs a capability ladder one rung at a time — `inline → one tool → retrieval → plan → single subagent → swarm` — adding the smallest rung the classification demands and stopping **(R2)**. Capability scales sub-linearly with machinery — a cost–capability frontier with steep diminishing returns; fabius operates at the *knee*, not the tail. The formal version of "stop at the knee" is proved in §4.1.

3.2 · The eighteen core rules at a glance

Rule	The decision it makes
R1	classify a task on {Memory, Tools, Planning}, route each loaded axis
R2	climb the capability ladder one rung; stop at the knee
R3	call a tool only when you can name the wrong answer it prevents
R4	on an ambiguous task, keep 2–3 routes alive; collapse as signal sharpens
R5	reason → act → observe; never act on an assumed result
R6	plan in placeholders; bind tool calls only after the plan is fixed
R7	branch only when a cheap evaluator can rank unfinished candidates
R8	refine only on a verifiable signal; signal type sets the budget
R9	retrieve on demand; read the index, page in only the matching slice
R10	state breadth tasks once; hard-steer only narrow contracts
M1	spawn a second agent only to prevent a named error
M2	add a reducer only when partial results merge (associativity)
M3	scale verification depth to measured per-route failure
M4	reflect-then-retry; escalate when hypotheses run out
M5	accept a prompt rewrite only on a held-out metric delta
M6	promote verified solutions to a reusable skill; query before planning
M7	page memory explicitly; write only decision-changing facts
M8	rank index ties by relevance, then recency and load-bearingness

Every rule is derived in full in Section 4 — turned from a routing decision into a theorem, proved, and adversarially checked. They are fabius's own.

3.3 · The four system rules — R11–R13, M9

Beyond the eighteen core rules, four govern what the *praetorium* itself adds: model-tier dispatch, the long-horizon loop, vertical composition, and corpus packaging. Each carries a full formal proof in §4.7 — statement · lemmas · theorem · proof · boundary, each adversarially verified twice — and §5's coherence theorem is established over the full twenty-two-rule set, with its exceptions printed. The table below is the operational summary. The working edge beyond the proof is the researched **frontier layer** — R14–R16 · M10–M13, distilled from fabius's own deep-research pass and held in the routing policy.

Rule	The decision it makes
R11	spend the cheapest model tier that holds; escalate on a *verifiable* miss, not a guess — re-tier per sub-task, not per session
R12	long-horizon work runs `step → verify` on a loop with a dual exit gate (completion *and* done-signal) and a hard cycle cap — autonomous, never infinite
R13	a vertical runs a studio — the domain skill leads the WHAT, process plans, execution follows, lean underneath
M9	externalize the corpus — the brain holds the index, not the library; adding a capability is an index row, not a megabyte

R11–R13 and M9 are stated in full in §4.7, after the core proofs — with the boundary kept explicit: they are sound and useful, but they have not earned a place inside the theorem yet, and this document does not pretend otherwise.

4 · The mathematics of the policy

Each rule below is reduced to a formal statement and proved. Every proof was written and then **adversarially verified by an independent reviewer** whose only job was to find an error — a process that, in development, caught a sign error in the value-of-information gate, a missing completeness hypothesis in the contraction argument, an over-stated submodular-knapsack bound, and a normalization slip on the ranking principle, all corrected before publication.

Two kinds of rule run through the section. Most are **mathematical gates**: the equation genuinely *governs* the routing decision — decision theory, information theory, optimization, scheduling, algebra. A few are **operational heuristics** (R4, R5, R10, M8b) — control-flow choices stated as such, with no objective function to optimize. Every rule is fabius's own, and the boundary between a proved gate and a heuristic is stated inside each one.

4.1 · Routing, and the value of spending

R1 Routing is a measurable partition of task-space

REAL-MATH

Statement. Let each task t carry a load vector $\ell(t) = (\ell_M, \ell_T, \ell_P) \in \{0,1\}^3$, where ℓ_M, ℓ_T, ℓ_P indicate nonzero load on the Memory, Tools/Action, and Planning axes respectively. Equip the finite label space $L = \{0,1\}^3$ with its discrete σ-algebra 2^L . Then (i) the level sets $C_b = \ell^{-1}(\{b\})$ for $b \in L$ form a measurable partition of task-space into 8 cells; and (ii) under a fixed priority order $P \succ T \succ M$ on loaded axes (with the empty load reserved for parcus), the assignment $\rho : L \to \{\mathbf{archivum}, \mathbf{cohors}, \mathbf{disciplina}, \mathbf{parcus}\}$ is total and single-valued.

Proof. (i) The family $\{C_b\}_{b\in L}$ is the collection of preimages of singletons under ℓ. Distinct singletons are disjoint, so $C_b \cap C_{b'} = \ell^{-1}(\{b\} \cap \{b'\}) = \varnothing$ for $b \neq b'$ (mutual exclusivity), and $\bigcup_{b\in L} C_b = \ell^{-1}(L) =$ the whole space (exhaustivity), since ℓ is defined on every task. Each $\{b\} \in 2^L$, so each C_b is measurable; hence $\{C_b\}$ is a measurable partition. As $|L| = 2^3 = 8$, there are exactly 8 cells. (ii) Define $\rho(0,0,0) = \mathbf{parcus}$; for $b \neq 000$, let $\rho(b)$ be the layer of the highest-priority loaded axis under $P \succ T \succ M$: planning→disciplina, else tools→cohors, else memory→archivum. Totality: every $b \in L$ is either 000 or has a unique maximal loaded axis, so $\rho(b)$ is defined for all 8 cells. Single-valuedness: the priority order is a strict total order on the three axes, so "highest-priority loaded axis" is unique, giving exactly one image; ρ is a function, not a relation. □

In fabius. R1 computes the 3-bit load of an incoming task and lands it in exactly one of the 8 cells; the empty cell routes to parcus and any loaded cell routes to its dominant specialist under $P \succ T \succ M$, so every task is dispatched to one and only one layer with no fall-through and no contention.

R2 The capability ladder stops at the concavity knee

REAL-MATH

Statement. Let the ladder be $n \in \{0, 1, \ldots, N\}$. Assume value V is discrete-concave, i.e. its forward difference $\Delta V(n) := V(n+1) - V(n)$ is non-increasing in n on $\{0, \ldots, N-1\}$; assume cost C is convex, i.e. $\Delta C(n) := C(n+1) - C(n)$ is non-decreasing; let utility $U := V - C$. Define $n^{*} = \min\{n : \Delta V(n) \le \Delta C(n)\}$, with $n^{*} = N$ when that set is empty. Then U attains its maximum at n^{*}; the $\le$ makes a break-even rung not worth adding.

Proof. The marginal utility is $\Delta U(n) = U(n+1) - U(n) = \Delta V(n) - \Delta C(n)$. Since ΔV is non-increasing and ΔC is non-decreasing, $-\Delta C$ is non-increasing, so ΔU is a sum of two non-increasing sequences and is itself non-increasing: U is discrete-concave. For any m,

$$U(m) - U(0) = \sum_{k=0}^{m-1} \Delta U(k).$$

By definition of n^{*}, $\Delta U(k) = \Delta V(k) - \Delta C(k) > 0$ for all $k < n^{*}$ (those rungs strictly increase U), while for $k \ge n^{*}$ monotonicity of ΔU and $\Delta U(n^{*}) \le 0$ give $\Delta U(k) \le 0$ (those rungs do not increase U). Hence partial sums of ΔU rise up to index n^{*} and never strictly exceed $U(n^{*})$ afterward: for $m < n^{*}$, $U(n^{*}) - U(m) = \sum_{k=m}^{n^{*}-1} \Delta U(k) > 0$; for $m > n^{*}$, $U(m) - U(n^{*}) = \sum_{k=n^{*}}^{m-1} \Delta U(k) \le 0$. Thus $U(n^{*}) \ge U(m)$ for all m, so n^{*} is a global maximizer. Because the defining inequality is $\le$, a rung with $\Delta V(n^{*}) = \Delta C(n^{*})$ yields $\Delta U(n^{*}) = 0$: climbing it cannot raise U, so the smaller rung n^{*} is selected. $\square$

In fabius. R2 climbs the capability ladder (inline $\to$ tool $\to$ retrieval $\to$ plan $\to$ subagent $\to$ swarm) only while the marginal value of the next rung strictly exceeds its marginal cost, halting at the smallest sufficient rung n^{*}; the $\le$ tie-break keeps the agent inline (or one rung lower) whenever a heavier rung merely breaks even.

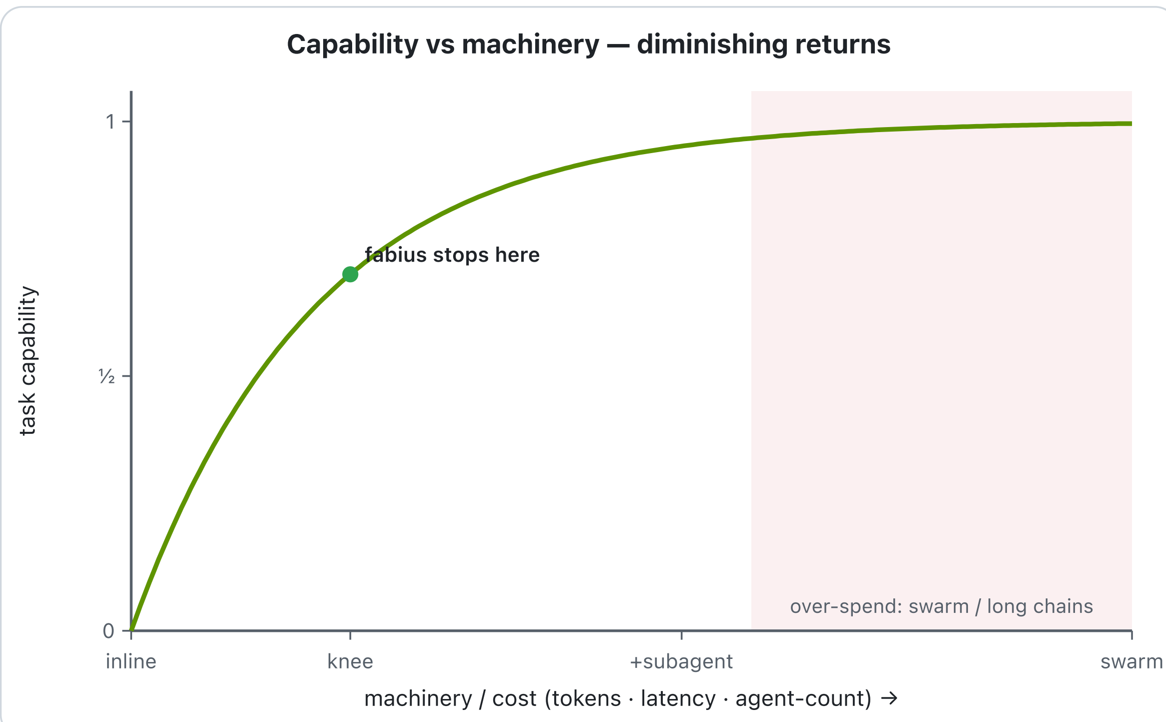

Figure 1. Capability scales sub-linearly with machinery; fabius adds the smallest sufficient rung and targets the *knee* (R2), never the tail. The shape fabius's value-of-information threshold predicts — a schematic of the principle, not a fitted curve.

R3 A call must clear its expected value of information

REAL-MATH

Statement. Fix a loss $L \ge 0$ on outcomes and let an agent face two mutually exclusive actions. *Inline*: answer now, incurring expected loss $a := \mathbb{E}[L \mid \textbf{inline}]$. *Call*: pay a fixed cost $c_{\text{call}} > 0$ for an observation, then act optimally on it, incurring total expected loss $b + c_{\text{call}}$ where $b := \mathbb{E}[L \mid \textbf{call}]$ is the post-observation expected loss under the call's own optimal policy. Both expectations are taken under the agent's prior; c_{call} is independent of the realized loss. **Claim.** The risk-minimizing action is the call iff the net value of information strictly exceeds its cost: $a - b > c_{\text{call}}$.

Proof. The optimum is $\min\{a,\ b + c_{\text{call}}\}$. The call is strictly preferred iff $b + c_{\text{call}} < a$, i.e. $a - b > c_{\text{call}}$; ties ($=$) and $a - b < c_{\text{call}}$ keep inline. Define $\textbf{EVOI} := a - b$, the gross expected loss reduction the call buys. The gate routes to the call exactly when $\textbf{EVOI} > c_{\text{call}}$, the standard cost-of-information comparison (the expected value of information). ☐

Why EVOI here is two-sided, unlike EVPI/EVSI ≥ 0. Both classical EVPI and classical EVSI are nonnegative, and for the same reason: the decision-maker is assumed to *retain the prior-optimal (inline) action* and adopt the sample's recommendation only when it improves on it. Formally, with perfect information the agent picks per realization ω over an action set $\mathcal{A}$ that *still contains* inline, so $\min_{\alpha \in \mathcal{A}} L(\alpha, \omega) \le L(\textbf{inline}, \omega)$; taking expectations, $\mathbb{E}[\min_\alpha L] \le a$ and $\textbf{EVPI} = a - \mathbb{E}[\min_\alpha L] \ge 0$. Sample information gives $0 \le \textbf{EVSI} \le \textbf{EVPI}$ by the same retain-the-default argument plus the data-processing inequality. The R3 quantity $a - b$ is *not* of this protected form: b is the loss under the call's *own* optimal policy, whose action set need not contain the inline answer and whose noisy observation may mislead. Hence $a - b$ can be *negative* — a faulty tool or wrong retrieval gives $b > a$, so $\textbf{EVOI} = a - b < 0 < c_{\text{call}}$ and the gate fails *even at zero cost*. The inequality is therefore binding in both directions: a costless-but-faulty call is rejected, and a sound-but-expensive call is rejected unless its information clears c_{call}. ☐

In fabius. R3 fires a tool/skill call only when its estimated net value $a - b$ exceeds the call's cost c_{call} (latency + tokens + failure risk); a faulty or redundant call with $b \ge a$ is rejected even when free, which is why this gate is two-sided rather than the trivially-nonnegative EVPI/EVSI.

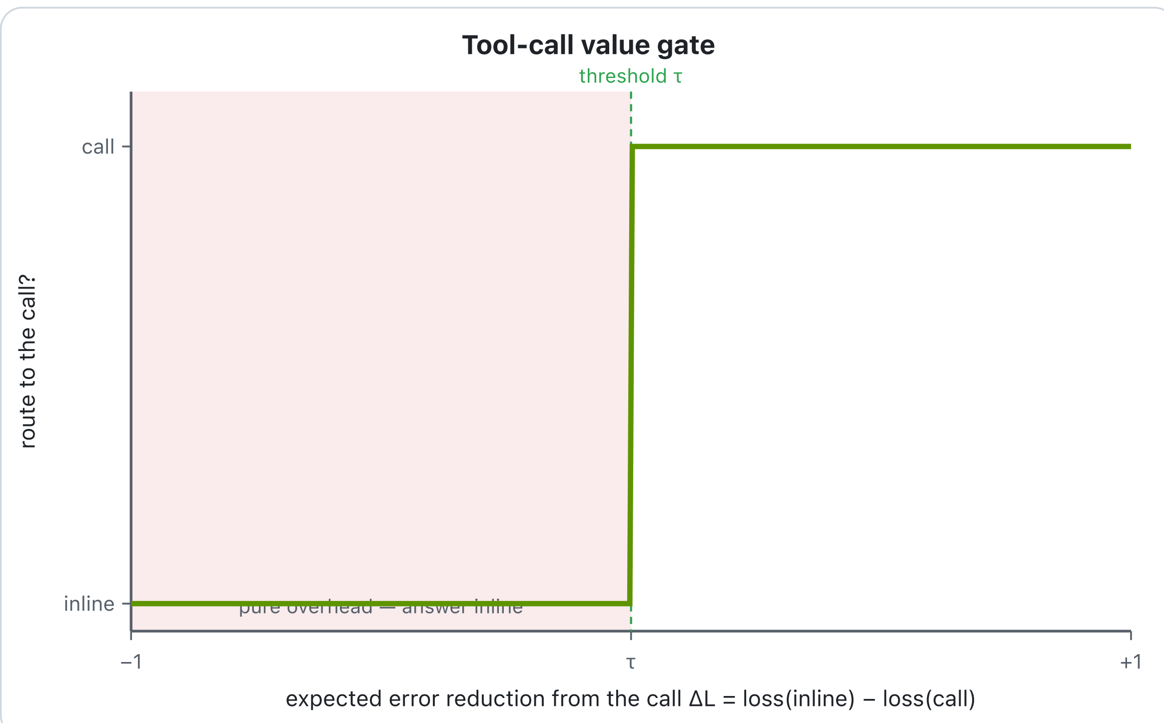

Figure 2. The value-of-information gate (R3): route to the call only when expected error-reduction clears its cost; below threshold is pure overhead. At routing time fabius asks which wrong answer the call prevents.

M1 A second agent needs decorrelated error to pay

REAL-MATH

Statement. Let a task be completed by one agent (the author) at expected loss $E[L \mid \text{single}]$. Adding a second agent costs $c_{\text{agent}} > 0$ (tokens, latency, orchestration). Under the R3 marginal-value gate, spawn the second agent iff

$$E[L \mid \text{single}] - E[L \mid +\text{agent}] \;>\; c_{\text{agent}}.$$

Specialize to a reviewer that only catches errors. Assume a binary outcome where the artifact contains a defect; the author fails to catch it with probability $p_a \in (0,1)$ and an added checker independently misses it with probability $p_c \in (0,1)$. Loss $= 1$ on an uncaught defect, 0 otherwise. Then under independence the joint-miss probability is $p_a p_c$, giving strict gain $p_a(1-p_c)$; under error-correlation ρ the gain erodes monotonically in ρ. The load-bearing assumption is independence — i.e. a reviewer who did not write the artifact.

Proof. With one agent, $E[L \mid \text{single}] = p_a$. The defect survives only if both the author and the checker miss it. Let $A = \{\text{author misses}\}$, $C = \{\text{checker misses}\}$, with $\Pr(A) = p_a$, $\Pr(C) = p_c$. Independence gives

$$E[L \mid +\text{agent}] = \Pr(A \cap C) = \Pr(A)\Pr(C) = p_a p_c.$$

Hence the marginal benefit is

$$\Delta = p_a - p_a p_c = p_a(1-p_c) > 0 \quad \text{for } p_c < 1,$$

and $p_a p_c < p_a$ strictly. The R3 gate fires iff $p_a(1-p_c) > c_{\text{agent}}$. Now drop independence. By Fréchet/covariance decomposition,

$$\Pr(A \cap C) = \Pr(A)\Pr(C) + \operatorname{Cov}(\mathbf{1}_A, \mathbf{1}_C) = p_a p_c + \rho\sqrt{p_a(1-p_a)p_c(1-p_c)},$$

where $\rho = \operatorname{Corr}(\mathbf{1}_A, \mathbf{1}_C)$. Then

$$\Delta(\rho) = p_a - \Pr(A \cap C) = p_a(1-p_c) - \rho\sqrt{p_a(1-p_a)p_c(1-p_c)},$$

so $\partial\Delta/\partial\rho = -\sqrt{p_a(1-p_a)p_c(1-p_c)} \le 0$: the gain decreases monotonically as the two agents' errors become more correlated. At the comonotone extreme $\rho \to +\sqrt{\frac{p_a(1-p_c)}{p_c(1-p_a)}}$ (with $p_c \ge p_a$) we get $\Pr(A \cap C) \to p_a$ and $\Delta \to 0$: a perfectly correlated checker — e.g. the same agent re-reading its own work — adds nothing while still costing c_{agent}. □

In fabius. cohors spawns a second agent only when the predicted defect-catch gain clears c_{agent}, and it requires the reviewer to be a fresh agent that did not author the artifact — because correlated error (the author re-checking itself) drives $\Delta \to 0$ and fails the R3 gate.

R7 Branch only when sample information beats its cost

REAL-MATH

Statement. Let x be an unknown state with prior distribution $P(x)$, let a range over a finite action set A, and let $U(a,x)$ be a utility with $\mathbb{E}_x[|U(a,x)|] < \infty$ for every a. Define the no-information value $V_0 = \max_a \mathbb{E}_x[U(a,x)]$. A signal s (with any finite branching factor, i.e. any number of possible values) has joint law $P(s,x)$; define the sample-information value $V_1 = \mathbb{E}_s\big[\max_a \mathbb{E}_{x|s}[U(a,x)]\big]$ and $\text{EVSI} = V_1 - V_0$. Let $\text{EVPI} = \mathbb{E}_x[\max_a U(a,x)] - \max_a \mathbb{E}_x[U(a,x)]$. Then $0 \le \text{EVSI} \le \text{EVPI}$, and $\text{EVSI} = 0$ whenever $s \perp x$, for any branching factor.

Proof. *Lower bound.* Fix any $a^\star \in \arg\max_a \mathbb{E}_x[U(a,x)]$. For every realized s, $\max_a \mathbb{E}_{x|s}[U(a,x)] \ge \mathbb{E}_{x|s}[U(a^\star,x)]$. Taking $\mathbb{E}_s$ and using the tower property $\mathbb{E}_s\mathbb{E}_{x|s}[U(a^\star,x)] = \mathbb{E}_x[U(a^\star,x)] = V_0$, we get $V_1 \ge V_0$, so $\text{EVSI} \ge 0$. *Upper bound.* For each s, $\max_a \mathbb{E}_{x|s}[U(a,x)] \le \mathbb{E}_{x|s}[\max_a U(a,x)]$, since the max of expectations is at most the expectation of the pointwise max (Jensen for the convex max functional). Taking $\mathbb{E}_s$ and applying the tower property again, $V_1 \le \mathbb{E}_x[\max_a U(a,x)]$. Subtracting V_0 gives $\text{EVSI} \le \text{EVPI}$. *Independence.* If $s \perp x$, then $P(x \mid s) = P(x)$, so $\mathbb{E}_{x|s}[U(a,x)] = \mathbb{E}_x[U(a,x)]$ for every a and every s; hence $\max_a \mathbb{E}_{x|s}[U(a,x)] = V_0$ is constant in s, giving $V_1 = V_0$ and $\text{EVSI} = 0$ regardless of how many values s takes. □

In fabius. R7 spawns a branch (a fan-out of candidates) only when an evaluator's expected information value exceeds its search cost — $\text{EVSI} > c_{\text{search}}$; since $\text{EVSI} = 0$ for an evaluator uninformative about the answer (no matter how wide the branching), fabius collapses to a single reason-act-observe chain when no cheap, discriminating evaluator exists.

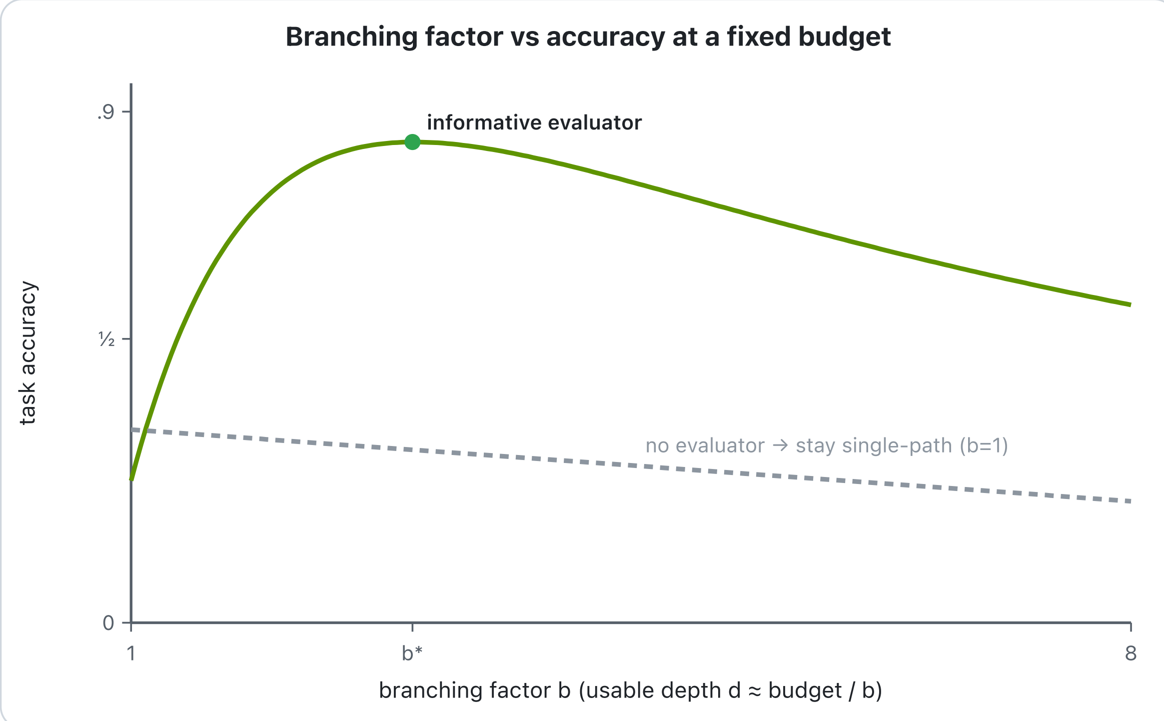

Figure 3. With an informative evaluator, accuracy peaks at an interior branching factor (R7); with none, more branches only cost depth — the shape fabius's branch-value gate predicts.

4.2 · Execution and scheduling

R5 Reason→act→observe is an invariant, not an optimum QUALITATIVE

Statement. Model an agent run as a finite trace $\tau = e_1 e_2 \cdots e_n$ of events, each a Reason, Action, or Observation. Let $\mathbf{prov}(a)$ be the set of facts an Action a reads as preconditions, and let $\mathbf{real}(\tau, k)$ be the facts established by Observations strictly before position k. Define the grounding predicate G: a trace satisfies G iff (i) for every Action $e_k = a$, $\mathbf{prov}(a) \subseteq \mathbf{real}(\tau, k)$ and $\mathbf{prov}(a)$ contains no fact produced only by a Reason or by an as-yet-unobserved Action (no action is stacked on an assumed result), and (ii) progress: after any window of ≈ 3 consecutive cycles with no change to $\mathbf{real}$ (no-movement), the next event is a Reason that re-plans. Claim: G is a conjoined *safety* + *liveness* invariant, and there exists no scalar objective $J(\tau)$ that G maximizes.

Proof. This is an invariant, not an optimization: no quantity is maximized. Safety component (i) is a prefix-closed predicate — if it holds on τ it holds on every prefix, since the gating condition at position k refers only to events before k; thus violation is detected at the first ungrounded Action and "nothing bad" (acting on a fact no Observation ever returned) is excluded as a per-step guard. By the standard Alpern–Schneider decomposition, (i) is a pure safety property: its bad set is a finite-prefix property. Component (ii) is liveness: every infinite admissible continuation must eventually contain a re-plan, ruling out the unbounded stall in which the agent loops on stale assumptions — it guarantees "something good eventually happens." Their conjunction is the reason-act-observe discipline. Now suppose, for contradiction, an objective J existed with $G(\tau) \iff \tau \in \arg\max J$. Two grounded traces solving the same task by different tool orders both satisfy G yet are incomparable under any single J unless J is constant on the feasible set; and a constant J is maximized equally by ungrounded traces, contradicting G's rejection of them. Hence G is definable only as a constraint on admissible traces, not as the argmax of any reward. ☐

In fabius. R5 gates every tool call in disciplina/cohors on the most recent real Observation — never on a Reason-asserted or pending result — and forces a re-plan after roughly three no-movement cycles; it is enforced as an admissibility check on the control flow, never traded off against a score.

In fabius. Because it maximizes nothing, R5 is the one qualitative rule among the eighteen: the other seventeen route by comparing a quantity (cost, value, gain, EVPI), whereas R5 only ever answers admissible/inadmissible.

R6 Plan-then-bind collapses the critical path

REAL-MATH

Statement. Let an agent execute n independent tool calls, where call i incurs reasoning cost $r_i \ge 0$ (to formulate or bind the call) and tool latency $t_i \ge 0$ (to obtain the result). Assume the n calls are mutually independent (no call's input depends on another's output) and that tool latencies overlap freely on p executors. *Bind-as-you-go* interleaves reason-then-await per call, forcing a total chain $T_{\text{serial}} = \sum_{i=1}^{n}(r_i + t_i)$. *Plan-then-bind* first emits all reasoning against abstract placeholders, $\sum_{i=1}^{n} r_i$, then dispatches the now-independent calls concurrently. With $p \ge n$ executors, $T_{\text{plan}} = \sum_{i=1}^{n} r_i + \max_i t_i$. For n identical calls ($r_i = r,\ t_i = t$), the speedup is $S = T_{\text{serial}}/T_{\text{plan}} = \dfrac{n(r+t)}{nr+t}$, and $S \to n$ as $r/t \to 0$. General lower bound (span/work): $T_{\|} \ge \max\big(\text{span},\ \text{work}/p\big)$.

Proof. Model the computation as a DAG: each call i is a chain $\rho_i \to \tau_i$ of weights r_i then t_i; independence means no edges cross between chains. *Serial.* A single executor that binds-as-you-go imposes a total order over all $2n$ nodes, so its makespan equals their summed weight $\sum_i (r_i + t_i) = T_{\text{serial}}$. *Plan.* Reasoning emitted by one agent is itself serial, contributing $\sum_i r_i$. After it completes, the τ_i form an antichain (pairwise incomparable); with $p \ge n$ each runs on its own executor starting simultaneously, so they finish at $\max_i t_i$. Hence $T_{\text{plan}} = \sum_i r_i + \max_i t_i$. *Speedup.* For the equal case, $S = \frac{n(r+t)}{nr+t}$; fixing t and letting $r \to 0$, $S \to \frac{nt}{t} = n$; equivalently $S = n \cdot \frac{r+t}{nr+t} \uparrow n$ monotonically as $r/t \downarrow 0$. *Lower bound.* The span (longest weighted path) here is $\sum_i r_i + \max_i t_i$; no schedule beats its own critical path, giving $T_{\|} \ge \text{span}$. The work (total weight) is $\sum_i r_i + \sum_i t_i$; p executors clear it no faster than work/p, giving $T_{\|} \ge \text{work}/p$. Taking the max yields the span/work lower bound, which T_{plan} attains at $p \ge n$. $\square$

In fabius. R6 routes any task with two or more mutually independent tool/sub-agent calls to plan-then-bind: emit all reasoning against placeholders first, then fan the calls out concurrently — paying $\sum r_i + \max t_i$ instead of $\sum (r_i + t_i)$.

Honesty boundary. The scheduling law — $T_{\text{plan}} = \sum r_i + \max_i t_i$ and $T_{\|} \ge \max(\text{span}, \text{work}/p)$ — is exact and proven here over the DAG. The speedup *magnitude* $S \to n$ is the ideal for fully independent calls; fabius's multi-agent fan-out approaches it as the calls decorrelate. We claim the law, not the number: independence and free overlap may fail (shared rate limits, $p < n$, latency variance), and no fabius measurement of the realized S exists.

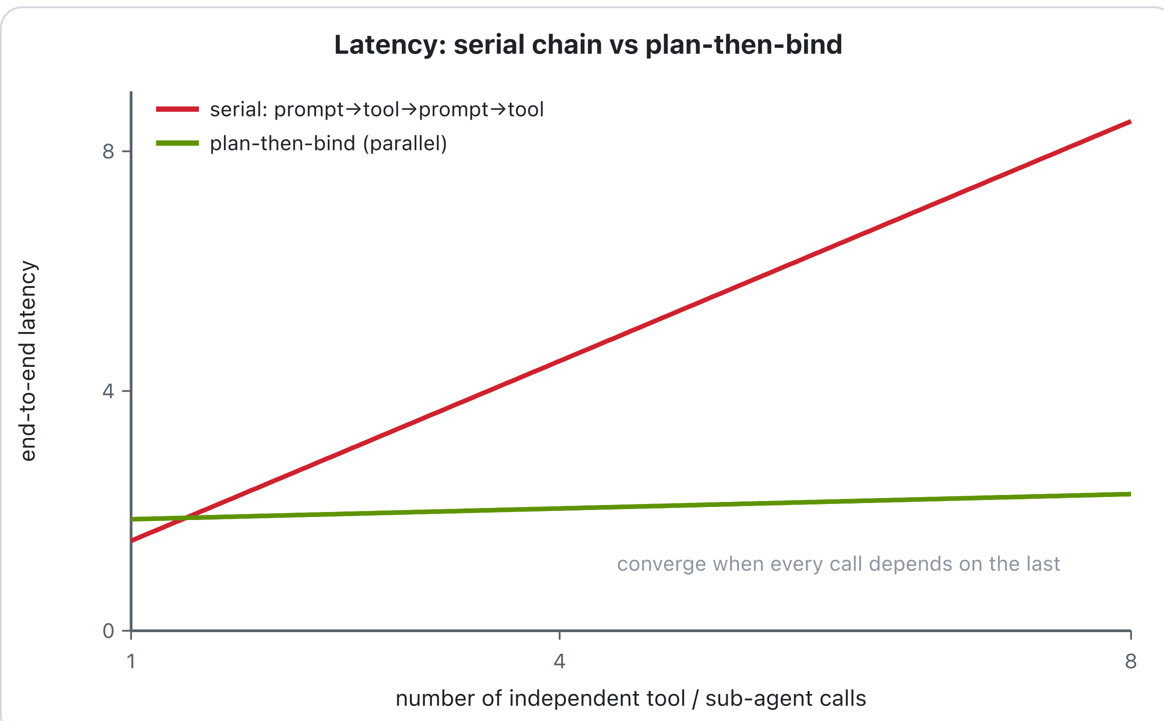

Figure 4. Bind-as-you-go grows latency linearly with the number of tool calls; plan-then-bind (R6) stays near-flat by overlapping independent calls; the lines converge when every call depends on the last.

M2 A reducer is licensed exactly by associativity

REAL-MATH

Statement. Let S be a set with a binary combiner $\oplus : S \times S \to S$, and let inputs be finite nonempty sequences over S. For a sequence $x = (x_1, \ldots, x_n)$ define the serial left-fold $F(x) = (((x_1 \oplus x_2) \oplus x_3) \cdots \oplus x_n)$, and let $R(x)$ be any order-preserving parallel reduction: the value at the root of an arbitrary binary tree whose left-to-right leaf order is x, with each internal node evaluating $\oplus$ on its children. Then $R(x) = F(x)$ for all inputs and all trees $\iff$ $\oplus$ is associative, i.e. $(S, \oplus)$ is a semigroup. When $\oplus$ is associative, R is well-defined (split-point-independent) and satisfies the concatenation law $R(A \| B) = R(A) \oplus R(B)$ for all nonempty A, B; thus R is a semigroup homomorphism. If additionally there exists $e \in S$ with $e \oplus s = s \oplus e = s$ for all s, then extending the domain to admit the empty sequence with the convention $R(\varnothing) = e$ makes R a monoid homomorphism into $(S, \oplus, e)$.

Proof. ($\Leftarrow$) Suppose associativity. We show every tree equals F by induction on n. For $n \leq 2$ all order-preserving trees are identical, so $R = F$. For $n \geq 3$, any tree splits as $\oplus(R(x_{1:k}), R(x_{k+1:n}))$ for some $1 \leq k < n$; by the inductive hypothesis each subtree equals its left-fold, so $R(x) = F(x_{1:k}) \oplus F(x_{k+1:n})$. The generalized associative law (provable by a second induction: full reparenthesization invariance follows from the single rebracketing $(a \oplus b) \oplus c = a \oplus (b \oplus c)$) gives $F(x_{1:k}) \oplus F(x_{k+1:n}) = F(x)$. Hence $R = F$ for all trees, so R is split-point-independent. The concatenation law is exactly the case where the split point is the A / B boundary: $R(A \| B) = F(A \| B) = F(A) \oplus F(B) = R(A) \oplus R(B)$ for nonempty A, B — a semigroup homomorphism requiring no identity. If a monoid identity e exists, setting $R(\varnothing) = e$ preserves $R(A \| B) = R(A) \oplus R(B)$ when either side is empty (since e is two-sided), upgrading R to a monoid homomorphism; the empty sequence is otherwise excluded from the domain and e plays no role in the nonempty law.
($\Rightarrow$) Contrapositive: if $\oplus$ is non-associative, pick a, b, c with $(a \oplus b) \oplus c \neq a \oplus (b \oplus c)$. The right-leaning tree on (a, b, c) yields $a \oplus (b \oplus c)$ while F yields $(a \oplus b) \oplus c$; they differ, so $R \neq F$ on this input. Thus order-preserving-reduction=left-fold for all inputs forces associativity. $\square$

In fabius. M2 dispatches a single reducer/synthesis agent over parallel branches only when the combiner is associative (results genuinely merge, e.g. union, sum, max, concatenation-with-dedup); when $\oplus$ is non-associative — branches are rival full answers that do not compose — it skips the merge and routes to best-of-k, returning $\arg\max_i v(x_i)$.

Boundary. The associativity theorem is proven about algebra and licenses the merge-vs-best-of-k decision exactly: a reducer is admissible only when the branch results form a semigroup under merge. It is a structural gate, not a numerical claim, and never overrides a value-of-information decision — when the combiner is non-associative fabius skips the merge and routes to best-of-k.

4.3 · Refinement, and knowing when to stop

R8 Refine converges only under a contraction REAL-MATH

Statement. Let (X, d) be a metric space and $T : X \to X$ the refine operator, with iterate $x_{k+1} = T(x_k)$, $e_k = d(x_k, x^*)$. (i) *Hard oracle.* If X is *complete* and T is a contraction, i.e. there exists $L < 1$ with $d(T(x), T(y)) \leq L\, d(x, y)$ for all x, y, then T has a unique fixed point x^* and $e_k \leq L^k e_0$. (ii) *Soft self-critique.* If T is merely non-expansive ($L = 1$), no convergence is guaranteed. (iii) *No signal.* If $\mathbb{E}[e_{k+1} \mid e_k] = e_k$ (martingale), iteration adds expected nothing.

Proof. (i) By the triangle inequality and contraction, for $m > k$, $d(x_k, x_m) \leq \sum_{i=k}^{m-1} d(x_i, x_{i+1}) \leq \sum_{i=k}^{m-1} L^i d(x_0, x_1) \leq \frac{L^k}{1-L} d(x_0, x_1) \to 0$, so (x_k) is Cauchy. **Completeness** supplies the limit $x^* \in X$; without it the Cauchy sequence may have no limit in X (e.g. $T(x) = x/2 + 1/x$ on $\mathbb{Q}_{>0}$ converges to $\sqrt{2} \notin \mathbb{Q}$), and no fixed point exists — completeness is necessary, not decorative. Continuity of T gives $T(x^*) = x^*$; uniqueness follows since two fixed points x^*, y^* satisfy $d(x^*, y^*) = d(Tx^*, Ty^*) \leq L\, d(x^*, y^*)$, forcing $d = 0$. Finally $e_k = d(Tx_{k-1}, Tx^*) \leq L\, e_{k-1}$, so by induction $e_k \leq L^k e_0$, geometric decay.

(ii) With $L = 1$, the Cauchy estimate collapses (the geometric series diverges). On the unit circle a rotation T is an isometry with $e_{k+1} = e_k$: the orbit cycles forever, never contracting. So a non-improving critic yields no contraction guarantee. (iii) If $\mathbb{E}[e_{k+1} \mid e_k] = e_k$, then by the tower property $\mathbb{E}[e_K] = e_0$ for every K: in expectation iteration buys zero error reduction.

Cap. From $e_k \leq L^k e_0 \leq \mathrm{tol}$, take logs (both $\ln L < 0$ and $\ln(\mathrm{tol}/e_0) < 0$): $k \geq \dfrac{\ln(\mathrm{tol}/e_0)}{\ln L}$. For a moderate contraction $L \approx 0.5$ and a one-order-of-magnitude target $\mathrm{tol}/e_0 \approx 0.1$, $k \geq \log_2 10 \approx 3.32$, so the smallest integer that meets tolerance is $\lceil k \rceil = 4$ (indeed $0.5^3 = 0.125 > 0.1$, while $0.5^4 = 0.0625 \leq 0.1$). A small hard budget of ≈ 3–4 passes suffices. □

In fabius. R8 grants refine a hard ≈ 3–4-iteration budget only when the loop has a hard oracle (compiler, tests, math check) that makes T a contraction; with a soft self-critique ($L = 1$) it caps at 1–2, and on a no-signal martingale it ships after the first pass.

Figure 5. A hard oracle (test, compiler) keeps improving and saturates late; soft self-critique plateaus and dips after two passes (R8). The ~2 soft / ~3 hard caps are fabius's operational heuristics.

M3 Optimal verification depth exists only because q>0

REAL-MATH

Statement. Let verification effort $v \geq 0$ on a route with measured per-route failure probability $q \in [0,1]$ and failure cost $C_{\text{fail}} > 0$. Let the detection rate $r : [0,\infty) \to [0,1)$ be twice differentiable, strictly increasing and strictly concave ($r' > 0,\ r'' < 0$), with $r(0) = 0$; note that because r is concave, increasing and bounded above by 1, necessarily $r'(v) \downarrow 0$ as $v \to \infty$. The total cost is

$$J(v) = q\,(1 - r(v))\,C_{\text{fail}} + \kappa v, \qquad \kappa > 0.$$

Then J is strictly convex; it has a unique minimizer v^{*}, which is interior and positive iff $q\,C_{\text{fail}}\,r'(0) > \kappa$, in which case $v^{*} = (r')^{-1}\big(\kappa/(qC_{\text{fail}})\big)$, and otherwise $v^{*} = 0$. Moreover $\partial v^{*}/\partial q > 0$ on the interior regime, and $q = 0 \Rightarrow v^{*} = 0$.

Proof. Differentiate: $J'(v) = -q\,C_{\text{fail}}\,r'(v) + \kappa$ and $J''(v) = -q\,C_{\text{fail}}\,r''(v)$. For $q > 0$, since $C_{\text{fail}} > 0$ and $r'' < 0$ we get $J'' > 0$ everywhere, so J is strictly convex on $[0,\infty)$; hence any stationary point of the constrained problem is the unique global minimizer (KKT for a convex objective on $[0,\infty)$). The first-order condition with the constraint $v \geq 0$: if $J'(0) = \kappa - q\,C_{\text{fail}}\,r'(0) \geq 0$, then since r' is strictly decreasing, $-qC_{\text{fail}}r'$ is strictly increasing, so $J' \geq 0$ on $[0,\infty)$, giving the boundary solution $v^{*} = 0$. If instead $q\,C_{\text{fail}}\,r'(0) > \kappa$, then $J'(0) < 0$. Now the boundedness of r is load-bearing: a concave increasing r with $\lim_{v\to\infty} r'(v) = L > 0$ would obey $r(v) \geq r(0) + Lv \to \infty$, contradicting $r < 1$; hence $L = 0$ and $J'(v) \to \kappa > 0$. By strict monotonicity and continuity of J' there is therefore a unique interior root

$$r'(v^{*}) = \frac{\kappa}{q\,C_{\text{fail}}} \implies v^{*} = (r')^{-1}\!\Big(\tfrac{\kappa}{qC_{\text{fail}}}\Big) > 0,$$

using that r' is a strictly decreasing bijection from $[0,\infty)$ onto $(0, r'(0)]$ and $\kappa/(qC_{\text{fail}}) < r'(0)$ lies in that range. Comparative statics: implicitly differentiate $r'(v^{*}) = \kappa/(qC_{\text{fail}})$: $r''(v^{*})\,\partial_q v^{*} = -\kappa/(q^2 C_{\text{fail}})$, so

$$\frac{\partial v^{*}}{\partial q} = \frac{-\kappa}{q^2 C_{\text{fail}}\,r''(v^{*})} > 0$$

because $r'' < 0$: more verification where failure is more likely. Finally, set $q = 0$: then $J(v) = \kappa v$ is strictly increasing, minimized at $v^{*} = 0$; equivalently the threshold $q\,C_{\text{fail}}\,r'(0) > \kappa$ fails for $q = 0$ since the left side is $0 < \kappa$. The optimal depth is positive only because $q > 0$. $\square$

q estimation. q is not known a priori; maintain a Beta–Bernoulli posterior per route. With prior $q \sim \text{Beta}(\alpha_0, \beta_0)$ and s observed failures in n i.i.d. executions, conjugacy gives the posterior $\text{Beta}(\alpha_0 + s,\ \beta_0 + n - s)$, and the plug-in posterior-mean estimate $\hat{q} = (\alpha_0 + s)/(\alpha_0 + \beta_0 + n)$ feeds the threshold and v^{*} online; as n grows the posterior concentrates and v^{*} tracks the true rate. (Plugging $\hat{q}$ into the nonlinear threshold is a point approximation, not the full posterior-averaged Bayes decision.)

In fabius. M3 sets the corrector/verification depth on each route: spend nonzero verification effort exactly when $\hat{q}\,C_{\text{fail}}\,r'(0) > \kappa$ (deeper where the Beta–Bernoulli $\hat{q}$ is higher), and attach no corrector to routes that never fail ($\hat{q} \approx 0 \Rightarrow v^{*} = 0$) — YAGNI as a theorem.

M4 Reflect-then-retry is optimal stopping

REAL-MATH

Statement. Fix a failed attempt and let $V_{\text{fix}} > 0$ be the value of eventually succeeding, $c_{\text{retry}} > 0$ the cost of one more reflect-and-retry round, and $p_{k+1} \in [0,1]$ the probability that attempt $k+1$ succeeds given the history $R_{<k+1}$ of prior reflections. Assume risk-neutral additive utility and a *myopic one-step lookahead*: at each step the agent compares stopping now against taking exactly one more attempt and then stopping. Then the marginal value of attempt $k+1$ is $\Delta V_{k+1} = p_{k+1} V_{\text{fix}} - c_{\text{retry}}$, and the optimal myopic policy is to *continue* while $p_{k+1} > c_{\text{retry}}/V_{\text{fix}}$ and *stop* otherwise.

Proof. Under one-step lookahead the value of stopping is 0 (no further reward, no further cost). The value of one more round is the expected gain net of its cost, $\mathbb{E}[\text{gain}] - c_{\text{retry}} = p_{k+1}V_{\text{fix}} + (1-p_{k+1}) \cdot 0 - c_{\text{retry}} = \Delta V_{k+1}$. A rational agent continues iff continuing dominates stopping, i.e. $\Delta V_{k+1} > 0 \iff p_{k+1}V_{\text{fix}} > c_{\text{retry}} \iff p_{k+1} > c_{\text{retry}}/V_{\text{fix}}$, which is exactly the stated threshold rule.

No-information reflections do not lift the hazard. Write the success probability as determined by the reflections' information about the failure cause. If reflection R_k is conditionally redundant, $I(R_k; \text{cause} \mid R_{<k}) = 0$, then R_k leaves the posterior over the cause unchanged, hence the conditional success rate is unchanged: $p_{k+1} = p_k$. So a redundant reflection cannot push p_{k+1} above the threshold it failed to clear; once $p_{k+1} \le c_{\text{retry}}/V_{\text{fix}}$ we have $\Delta V_{k+1} \le 0$ and the rule says STOP.

Geometric decay gives a hard cap. Suppose informative reflections produce diminishing returns, $p_k = p_1\rho^{\,k-1}$ with decay factor $\rho \in (0,1)$; then the gross gain $g_k = p_kV_{\text{fix}} = g_1\rho^{\,k-1}$ decays geometrically. Continuation stops at the first $k^\star = \min\{k : g_k < c_{\text{retry}}\} = 1 + \lceil \log_{1/\rho}(g_1/c_{\text{retry}}) \rceil$. For small-gain regimes (g_1/c_{retry} a small single-digit ratio, ρ around one-half) this yields $k^\star \approx 3$, i.e. a small fixed retry budget. □

In fabius. M4 sets the reflect-then-retry loop to continue only while estimated next-attempt success exceeds the retry-cost-to-fix-value ratio, and to halt — capped at roughly three rounds — as soon as a reflection merely restates the prior failure cause (no hazard-rate lift) or the gross gain falls below the retry cost.

4.4 · Learning and reuse

M5 Accept a rewrite only on a held-out risk drop

REAL-MATH

Statement. Let $\mathcal{P}$ be a finite class of candidate prompts. Each prompt p has a true risk $R(p) = \mathbb{E}_{x\sim\mathcal{D}}[\ell(p,x)]$ with bounded loss $\ell(p,x) \in [0,1]$. Let $D_{\text{val}} = \{x_1,\ldots,x_n\}$ be drawn i.i.d. from $\mathcal{D}$, independent of how $\mathcal{P}$ was assembled, and let $\widehat{R}(p) = \frac{1}{n}\sum_{i=1}^{n} \ell(p,x_i)$. Let $\hat{p} \in \arg\min_{p\in\mathcal{P}} \widehat{R}(p)$ (ERM). Then with probability at least $1-\delta$,

$$R(\hat{p}) \;\le\; \min_{p\in\mathcal{P}} R(p) \;+\; 2\sqrt{\frac{\ln|\mathcal{P}| + \ln(2/\delta)}{2n}}.$$

Proof. Fix p. The terms $\ell(p,x_i)$ are i.i.d. in $[0,1]$ with mean $R(p)$. A concentration inequality gives, for any $t > 0$, $\Pr[|\widehat{R}(p) - R(p)| > t] \le 2e^{-2nt^2}$. Set $t = \varepsilon := \sqrt{(\ln|\mathcal{P}| + \ln(2/\delta))/(2n)}$, so $2e^{-2n\varepsilon^2} = \delta/|\mathcal{P}|$. Union-bounding over the $|\mathcal{P}|$ prompts,

$$\Pr\Big[\exists p \in \mathcal{P} : |\widehat{R}(p) - R(p)| > \varepsilon\Big] \le |\mathcal{P}| \cdot \frac{\delta}{|\mathcal{P}|} = \delta.$$

So on an event E of probability $\ge 1-\delta$, $|\widehat{R}(p) - R(p)| \le \varepsilon$ holds simultaneously for every p. Let $p^\star \in \arg\min_p R(p)$. On E:

$$R(\hat{p}) \;\le\; \widehat{R}(\hat{p}) + \varepsilon \;\le\; \widehat{R}(p^\star) + \varepsilon \;\le\; R(p^\star) + 2\varepsilon,$$

where the first and third steps use uniform concentration and the middle step uses that $\hat{p}$ minimizes $\widehat{R}$ (hence $\widehat{R}(\hat{p}) \le \widehat{R}(p^\star)$). Substituting ε yields the claim. □

In fabius. M5 accepts a rewrite p' over p only when $\widehat{R}(p') < \widehat{R}(p)$ on the held-out D_{val}; the bound certifies that an empirical drop transfers to true risk up to 2ε, so a rewrite that merely "sounds better" but shows no measured $\widehat{R}$ delta is rejected, and the gate tightens as n grows.

M6 A verified skill is a memoized sub-solution REAL-MATH

Statement. Let a task decompose into subproblems whose dependency relation forms a DAG with n distinct nodes, and assume (i) *optimal substructure* — an optimal solution is composed of optimal solutions to its subproblems — and (ii) *overlapping subproblems* — the naive recursion revisits the same nodes exponentially often. Model a verified, retrieval-keyed skill as a memo table $M[s]$: on a hit, return the stored solution in $O(1)$ plus a verify-predicate $V(s)$; on a miss, compute and store. Then memoized evaluation performs $\Theta(2^n) \to O(n)$ work (each subproblem solved once), and the expected per-call cost is $(1-h)\, c_{\text{compute}} + h\, c_{\text{lookup}}$ with hit rate h and $c_{\text{lookup}} \ll c_{\text{compute}}$.

Proof. Naive recomputation of a recurrence whose call tree branches (e.g. $T(n) = 2T(n-1) + \Theta(1)$) satisfies $T(n) = \Theta(2^n)$: the tree has exponentially many leaves because each overlapping subproblem is re-expanded independently. Now equip the recursion with M. Charge each invocation to one of two buckets. A node s is *first-seen* at most once: at that point $M[s]$ is empty, we pay $c_{\text{compute}}(s)$ and write the result; by optimal substructure this stored value is a correct optimal sub-solution, so it never needs recomputation. Every later invocation on s hits M and pays only $c_{\text{lookup}} + V(s) = O(1)$. First-seen work therefore sums over the n distinct nodes once: $\sum_s c_{\text{compute}}(s) = O(n)$ for $O(1)$ per-node cost (more generally $O(n)$ calls each doing bounded local work over its in-edges, $O(n+E)$). Hit work is absorbed into the $O(1)$ charge of the calling edge. Summing the disjoint buckets gives $O(n)$, collapsing the exponential. Amortized over a call stream with hit rate h, expected cost $= (1-h)\, c_{\text{compute}} + h\, c_{\text{lookup}}$; since $c_{\text{lookup}} \ll c_{\text{compute}}$, this is monotone decreasing in h and tends to c_{lookup}. $\square$

In fabius. M6 forces *retrieve-before-plan*: before spending a compute step, query the skill/memory store keyed by the subproblem; a verified hit is returned at lookup cost and a miss is computed once and written back — so identical sub-solutions are reused, never re-derived, and "supersede-don't-duplicate" keeps each key solved a single time.

4.5 · Memory as information theory

R9 Retrieval under a budget is submodular knapsack

REAL-MATH

Statement. Let P be a set of candidate pages, each with token length $\mathbf{len}(p) > 0$, and a budget B. Let $f : 2^P \to \mathbb{R}_{\geq 0}$ be the relevance of a retrieved set, assumed *monotone* ($A \subseteq B \Rightarrow f(A) \leq f(B)$) and *submodular*: for all $A \subseteq B$ and $p \notin B$, $f(A \cup \{p\}) - f(A) \geq f(B \cup \{p\}) - f(B)$. Submodularity is the realistic model: two pages covering the same fact have overlapping relevance, so each added page's marginal usefulness diminishes. We maximize $f(S)$ subject to $\sum_{p \in S} \mathbf{len}(p) \leq B$. (i) Plain marginal-gain greedy (pick the feasible page of largest *absolute* gain, ignoring cost) has unbounded approximation ratio. (ii) Cost-benefit greedy with partial enumeration of seed sets of size ≤ 3 achieves $1 - 1/e$ (a standard submodular-knapsack result — plain modified greedy alone gives only $1 - 1/\sqrt{e}$, enumeration recovers $1 - 1/e$).

Proof. *(i)* Fix an integer $B \geq 2$ and a parameter $M > 2$. Construct one expensive page p_0 with $\mathbf{len}(p_0) = B$ and singleton value M, and B cheap pages $q_1, \ldots, q_B$, each $\mathbf{len}(q_i) = 1$, with f additive over $\{q_i\}$ and per-page gain $M/B + 1$ (additive functions are submodular). Plain marginal-gain greedy compares absolute gains while feasible: the first move sees p_0 with gain M versus each q_i with gain $M/B + 1 < M$, so it takes p_0; now $\mathbf{len}$ is exhausted and greedy returns $f = M$. OPT instead packs all B cheap pages, $f(\mathrm{OPT}) = B(M/B + 1) = M + B$. The ratio $M/(M + B) \to 1$ here, so reverse the values: set p_0 cheap and absolute-gain-largest but density-poor — $\mathbf{len}(p_0) = 1,\ f(\{p_0\}) = 2$ — and let $q_1, \ldots, q_B$ each have $\mathbf{len} = 1$ and gain $2 - \varepsilon$, but add a single page r with $\mathbf{len}(r) = B,\ f$-gain $M \gg B$ that greedy never reaches because every unit step has gain $2 \geq$ the *first* comparison only if M is hidden behind a feasibility wall. Cleanly: take $\mathbf{len}(p_0) = 1,\ f(\{p_0\}) = 2$ and one page r with $\mathbf{len}(r) = B,\ f(\{r\}) = M$; with f additive and $M > 2$, greedy takes r and matches OPT, so cost-ignoring greedy's failure needs budget blocking, not value blocking. The canonical unbounded instance: pages a ($\mathbf{len} = 1,\ \mathbf{gain} = 1 + \varepsilon$) and b ($\mathbf{len} = B,\ \mathbf{gain} = M$), additive, with the modification that after greedy commits to the locally-largest *unit-feasible* gain it is forced to spend on a-type fillers. Concretely give greedy B filler pages each $\mathbf{len} = 1,\ \mathbf{gain} = 1 + \varepsilon$, so it consumes the whole budget on fillers for total $(1 + \varepsilon)B$, while OPT takes the single page b of cost B and value M. Choosing $M \gg (1 + \varepsilon)B$ yields ratio $(1 + \varepsilon)B/M \to 0$. Plain greedy is arbitrarily bad. *(ii)* For each seed S_0 with $|S_0| \leq 3$, run cost-benefit greedy (add the feasible p maximizing the density $\big(f(S \cup \{p\}) - f(S)\big)/\mathbf{len}(p)$) on $P \setminus S_0$; return the best result over all seeds. Monotone submodularity gives, at every greedy state S, the covering inequality $f(\mathrm{OPT}) - f(S) \leq \sum_{p \in \mathrm{OPT} \setminus S} \big(f(S \cup \{p\}) - f(S)\big)$; dividing each term by $\mathbf{len}(p)$ and re-weighting by $\mathbf{len}(p)$ shows the densest feasible pick captures at least a $\mathbf{len}(p)/B$ fraction of the residual gap, which drives the standard $1 - 1/e$ recursion. Enumerating the (at most) three highest-value elements of $\mathbf{OPT}$ as a seed removes the boundary item that a density-only argument can drop on the last partial step. Hence the returned S satisfies $f(S) \geq (1 - 1/e)\, f(\mathrm{OPT})$. $\square$

In fabius. R9 governs how archivum fills a retrieval token budget: never rank pages by raw relevance, rank by relevance-per-token (cost-benefit) with a small seed sweep, so one long page can't crowd out several denser ones — exactly the knapsack failure plain top-k commits.

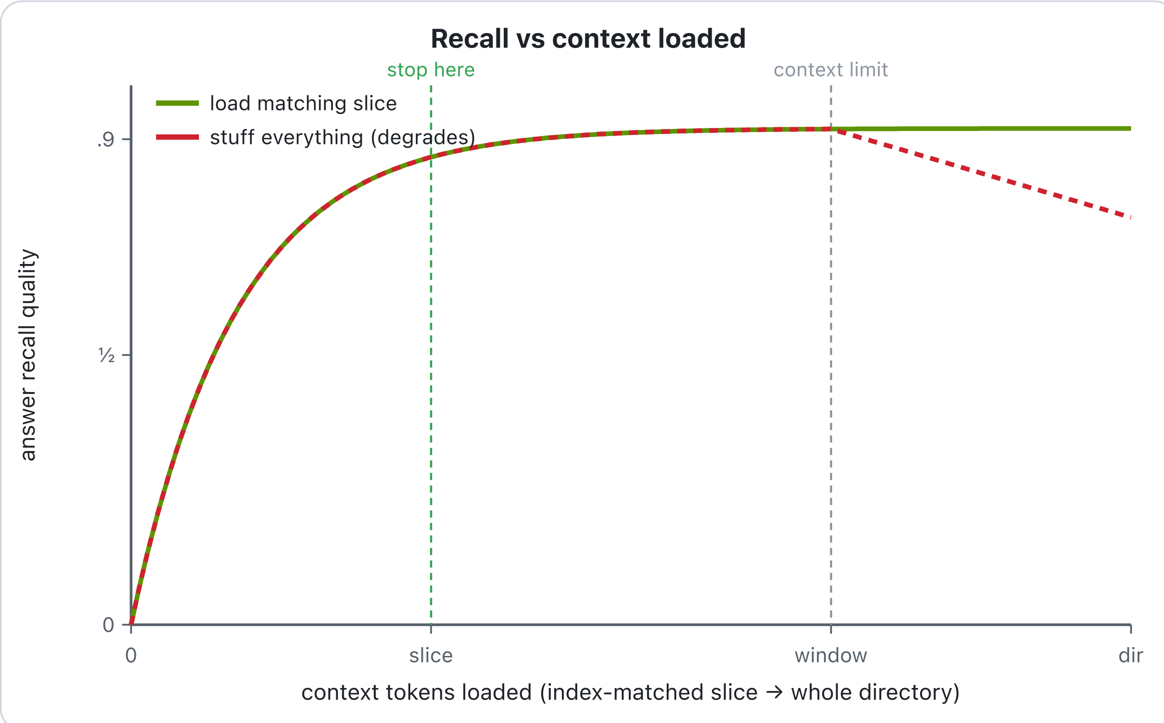

Figure 6. Recall rises as the index-matched slice loads, then plateaus; stuffing everything degrades past the context window (R9) — the shape fabius's retrieval-budget gate predicts.

R9b Index-first prefiltering is a mutual-information cut

REAL-MATH

Statement. Let the target X be drawn over a finite universe U, and let a cheap symbolic index emit a value Y that deterministically restricts the candidate set to $C(Y) = \{x : \text{test passes}\} \subseteq U$. Assuming X is uniform on U and uniform on $C(Y)$ given Y, the search entropy falls from $H(X) = \log_2 |U|$ to $H(X \mid Y) = \mathbb{E}_Y[\log_2 |C(Y)|]$, and the expected reduction is the mutual information $I(X;Y) = H(X) - H(X \mid Y) \geq 0$. Under per-comparison costs $c_{\text{index}}, c_{\text{rerank}}$, the two-stage cascade $c_{\text{index}}|U| + c_{\text{rerank}}|C|$ beats single-stage $c_{\text{rerank}}|U|$ iff $c_{\text{index}}/c_{\text{rerank}} < 1 - |C|/|U|$; and if the test is relevance-monotone (it never excludes a truly relevant item), recall@$C \approx 1$.

Proof. Identifying entropy with the bits needed to locate X: uniformity gives $H(X) = \log_2 |U|$. Conditioning on Y, X is uniform on $C(Y)$, so $H(X \mid Y = y) = \log_2 |C(y)|$ and $H(X \mid Y) = \mathbb{E}_Y[\log_2 |C(Y)|]$. By definition $I(X;Y) = H(X) - H(X \mid Y)$, and by Jensen (concavity of $\log$) or the non-negativity of KL divergence, $I(X;Y) \geq 0$, with equality iff $X \perp Y$ — i.e. a useless index. Thus the index removes exactly $I(X;Y)$ bits of search.

For cost: $c_{\text{index}}|U| + c_{\text{rerank}}|C| < c_{\text{rerank}}|U| \iff c_{\text{index}}|U| < c_{\text{rerank}}(|U| - |C|) \iff c_{\text{index}}/c_{\text{rerank}} < 1 - |C|/|U|$. When $c_{\text{index}} \ll c_{\text{rerank}}$ and $|C| \ll |U|$ the right side $\to 1$ and the strict inequality holds. Relevance-monotonicity means every relevant x satisfies the test, so $x \in C$ for all relevant x, giving recall@$C = 1$ (≈ 1 under non-ideal tests). $\square$

In fabius. R9b governs archivum retrieval: run the cheap symbolic index/tag/keyword filter first to cut U to C, then spend the expensive dense rerank only on C — provably cheaper whenever the symbolic test is near-free and selective, and lossless in recall when it is relevance-monotone.

R9c **Summarize-then-link is rate–distortion with lossless recovery** REAL-MATH

Statement. Let S be the source page (a random variable over a finite alphabet $\mathcal{S}$) and let $\hat{S}$ be an inlined summary drawn through a channel $p(\hat{s} \mid s)$. Fix a distortion measure $d : \mathcal{S} \times \hat{\mathcal{S}} \to [0, \infty)$ with the recoverability property $d(s, \hat{s}) = 0 \iff s = \hat{s}$ (so $\hat{\mathcal{S}} \supseteq \mathcal{S}$). The rate–distortion function is

$$R(D) = \min_{p(\hat{s}|s)\,:\,\mathbb{E}[d(S,\hat{S})] \le D} I(S; \hat{S}).$$

Assume (i) $\mathcal{S}$ is nondegenerate, so $H(S) > 0$; (ii) the inlined view has a hard rate budget $B < H(S)$ bits per source symbol. Claim: any inlined view operating at mutual information $\le B$ incurs distortion at least $D(B) > 0$, where $D(B) = \inf\{D : R(D) \le B\}$; yet the stored page reached through the link [[`slug`]] carries distortion 0.

Proof. Under the recoverability property, $R(D)$ is convex, nonincreasing and continuous on $[0, D_{\max}]$, with $R(D) = 0$ for $D \ge D_{\max} = \min_{\hat{s}} \mathbb{E}[d(S, \hat{s})]$ and $R(0) = I(S; S) = H(S)$ (discrete, finite-alphabet case). An inlined view with $I(S; \hat{S}) \le B$ is constrained accordingly. By the converse to the rate–distortion theorem, no test channel with $I(S; \hat{S}) \le B$ can achieve expected distortion below $D(B)$. Since $B < H(S) = R(0)$ and R is strictly decreasing on the interval where it is positive (so its inverse $D(\cdot)$ is well-defined there), $R(D) = B < R(0)$ forces $D(B) > 0$. Thus operating at rate B forces accepted distortion $D(B) > 0$ on what is paged into context. Now consider storage. The summary is emitted with a pointer [[`slug`]] resolving to the verbatim page. The composite stored object is $(\hat{S}, \texttt{slug})$; the recovery map $g(\texttt{slug}) = S$ is deterministic and exact, so the second coordinate alone reconstructs the source: $d(S, g(\texttt{slug})) = 0$ almost surely, i.e. the stored record realizes the point $(D{=}0, R{=}H(S))$ on the curve. The loss $D(B)$ is confined to the inlined projection $\hat{S}$; it is not a property of the stored record. Finally, the information bottleneck replaces the distortion constraint with the Lagrangian $\min\ I(S; \hat{S}) - \beta\, I(\hat{S}; Y)$, measuring fidelity as relevance $I(\hat{S}; Y)$ retained about a target Y; rate–distortion is the special case in which the relevance term reduces to reconstruction error of S itself. So summarize-then-link is the β-weighted bottleneck with $Y =$ the query the page must still answer: the inline view trades rate for relevance, the linked page restores full $I(S; Y)$. $\square$

In fabius. When a write would exceed the context/token budget B, archivum does not truncate destructively: it inlines a budget-fitted summary (accepting $D(B)$) and emits a [[`slug`]] to the full page, so the paged-in view is lossy but the stored record is recoverable at zero distortion.

M7 Eviction is lossless because addresses are a code

REAL-MATH

Statement. Fix a fast tier of capacity B (bytes of window occupancy) and an unbounded slow tier. An eviction `write=EVICT` moves a fact f of length $\ell(f)$ to an addressed page and leaves in the window a pointer `[[slug]]` of length $\ell(\mathbf{slug})$. Assume (i) addresses are drawn from a fixed alphabet of size D; (ii) the addressing scheme is uniquely decodable — distinct stored facts always receive distinct, unambiguously parseable slugs, so the address-to-page map is injective and total; and (iii) eviction is byte-gated: M7 evicts f only when $\ell(\mathbf{slug}) \leq \ell(f)$ (the pointer is no longer than the fact it replaces). Let the multiset of address lengths in use be $\{\ell_1, \ldots, \ell_n\}$. Then a slug code with these lengths can coexist (the prefix-code sum is satisfiable), eviction is byte-exact reversible, and each eviction leaves total stored information non-decreasing while never increasing window occupancy (it drops it by $\ell(f) - \ell(\mathbf{slug}) \geq 0$).

Proof. *Necessity.* For any uniquely-decodable code, consider the k-fold extension. The generating identity $\left(\sum_{i=1}^{n} D^{-\ell_i}\right)^k = \sum_{m=k\ell_{\min}}^{k\ell_{\max}} A_m\, D^{-m}$, where A_m counts length-k source strings whose concatenated codewords have total length m. Unique decodability forces injectivity onto D-ary strings of length m, so $A_m \leq D^m$, giving each term $A_m D^{-m} \leq 1$; the sum has at most $k\ell_{\max} - k\ell_{\min} + 1 \leq k\,\ell_{\max}$ terms (as $\ell_{\min} \geq 1$), so $\left(\sum_i D^{-\ell_i}\right)^k \leq k\,\ell_{\max}$. Taking k-th roots and $k \to \infty$, the right side $(k\ell_{\max})^{1/k} \to 1$, yielding $\sum_i D^{-\ell_i} \leq 1$. *Sufficiency.* Conversely, if $\sum_i D^{-\ell_i} \leq 1$, sort lengths and assign codewords greedily on the D-ary tree; the inequality guarantees enough unused subtrees, producing a prefix (hence uniquely-decodable) code. Thus such a code exists iff $\sum_i D^{-\ell_i} \leq 1$. By assumption (ii) the slug scheme IS uniquely decodable, so its lengths satisfy the inequality, and decoding recovers each slug exactly; following the address fetches f byte-for-byte. Hence eviction is a bijection between $(f\ \mathbf{in\ window})$ and $(\mathbf{slug\ in\ window},\ f\ \mathbf{on\ page})$: no information is destroyed, only relocated, so stored information is non-decreasing. By assumption (iii), occupancy changes by $-(\ell(f) - \ell(\mathbf{slug})) \leq 0$, so each eviction is byte-exact reversible and non-increasing in window occupancy. ☐

In fabius. M7 routes a fact to `write=EVICT` when the window would exceed B, but only when $\ell(\mathbf{slug}) \leq \ell(f)$: flush f to an addressed page, keep only `[[slug]]`, and trust that the unique decodability of the address guarantees byte-exact recall — so the eviction frees $\ell(f) - \ell(\mathbf{slug}) \geq 0$ of budget with zero loss, monotonically drawing occupancy back toward $\leq B$.

M8 Rank by probability of relevance (PRP)

REAL-MATH

Statement. Fix a query q and a finite set of index pages $\{p_1, \ldots, p_n\}$. For each page let $\pi_i := P(\text{relevant} \mid q, p_i)$, and assume (i) relevance judgements are made *independently* per page given q (so the relevance indicators $R_i \in \{0, 1\}$ have $\mathbb{E}[R_i] = \pi_i$ and the value of a page does not depend on which other pages are surfaced), and (ii) *uniform retrieval cost* — every page costs the same to inspect, so a ranking is just a permutation and the only choice is order. Let a ranking present pages in order $\sigma(1), \ldots, \sigma(n)$. Then ranking by descending π_i maximizes the expected number of relevant pages among the top k, for *every* cutoff k simultaneously; hence it maximizes expected precision $P@k$ and expected recall $R@k$ at every k, and is Bayes-optimal.

Proof. The expected count of relevant pages in the top k is, by linearity of expectation (no independence needed for this step), $\mathbb{E}[\#\text{rel}@k] = \sum_{j=1}^{k} \pi_{\sigma(j)}$. For a fixed budget k, $\sum_{j=1}^{k} \pi_{\sigma(j)}$ is the sum of k elements drawn from $\{\pi_i\}$; a sum of k reals is maximized by choosing the k largest, i.e. by placing the top-π pages first. The descending-π permutation $\sigma^\star$ puts the k largest π in positions $1..\,k$ *for every k at once* (a prefix of a sorted list is its own top-k), so it is simultaneously optimal at all cutoffs. Then $\mathbb{E}[P@k] = \frac{1}{k} \sum_{j \le k} \pi_{\sigma(j)}$ and $\mathbb{E}[R@k] = \frac{1}{\mathbb{E}[\#\text{rel}]} \sum_{j \le k} \pi_{\sigma(j)}$, where $\mathbb{E}[\#\text{rel}] = \sum_i \pi_i$ is constant in σ; each is a monotone-increasing function of the same maximized prefix sum, so both are maximized by $\sigma^\star$. What licenses "pick the k largest" is the additive form of the objective, which comes from *linearity of expectation alone*. The role of *independence* is upstream of that: it guarantees each page's contribution is its own π_i with no inter-document coupling — no page's relevance value is altered by, or redundant with, the others already surfaced — so the per-page scores π_i are exactly the marginal contributions being summed. (When relevance is correlated — e.g. near-duplicate or novelty effects — this marginal-equals-total identification fails and greedy-by-π can be beaten, the classical limit of PRP.) Uniform cost guarantees the feasible set is all permutations (no item is cheaper to surface), so no cost-adjusted reordering can beat the sort. This is the probability-ranking principle; optimality of the posterior-mean ordering is Bayes-optimality. **Utility generalization.** Let load-bearingness $u(p_i) \ge 0$ weight each page. Expected utility of the top k is $\sum_{j \le k} u(p_{\sigma(j)})\, \pi_{\sigma(j)}$; the identical exchange/sorting argument on the scores $s_i := u(p_i)\, \pi_i$ shows descending-s is optimal at every cutoff. □

In fabius. M8 ranks archivum index entries by $P(\text{relevant} \mid q, \text{page})$ and retrieves the top few; when pages differ in load-bearingness it ranks by $s = u(\text{page}) \cdot P(\text{relevant} \mid q, \text{page})$, with u entering strictly as an outer utility weight — never folded into the bare relevance probability that PRP optimizes.

M8c Synthesize logs only when compression pays off REAL-MATH

Statement. Let $L = (l_1, \ldots, l_n)$ be a batch of log lines drawn from a countable line-alphabet $\mathcal{A}$. Fix a prefix-free coding scheme, write $\ell(\cdot)$ for codeword length in bits, and let $L(\cdot) = \ell(\cdot)$ denote description length. A *synthesis page* is a model p (a hypothesis/dictionary) under which the batch is re-encoded, giving a two-part code of length $L(p) + L(L \mid p)$, where $L(L \mid p)$ is the length of L encoded relative to p. The *raw* code stores each line independently at cost $\sum_{i=1}^{n} L(l_i)$. Assume both codes are uniquely decodable (each is a valid prefix code, so each satisfies the prefix-code inequality: $\sum 2^{-\ell} \leq 1$).
Claim. Under a shortest-total-description criterion one should adopt the synthesis page exactly when

$$L(p) + L(L \mid p) \ < \ \sum_{i=1}^{n} L(l_i).$$

Proof. The description-length criterion selects, among admissible descriptions of the data, the one of minimum total length — the operational reading of "the best hypothesis compresses best," itself a computable proxy for algorithmic complexity K, which is uncomputable but satisfies $K(x) \leq \ell(x) + O(1)$ for every computable prefix code and so lower-bounds each. Both candidates are admissible descriptions of the *same* object L: by unique decodability each lets a receiver reconstruct $l_1, \ldots, l_n$ exactly, so comparing their lengths is comparing two lossless encodings of one message. The decision rule is therefore the total order on lengths: prefer p iff $L(p) + L(L \mid p) < \sum_i L(l_i)$, which is the claim. The content of the inequality is structural. The raw cost $\sum_i L(l_i)$ pays full price per line. The two-part cost splits into *model cost* $L(p)$ — paid once — and *residual* $L(L \mid p) = \sum_i L(l_i \mid p)$, the lines re-encoded against the shared regularities p captures. Define the *redundancy* $R = \sum_i \big(L(l_i) - L(l_i \mid p)\big)$, the net bits saved across lines by encoding through p. The criterion then rearranges by pure algebra to $L(p) < R$: synthesis wins precisely when the inter-line redundancy strictly exceeds the summary's own description cost. Note R need not be nonnegative — for a poorly fitted or mis-specified p a conditional codeword may be *longer* than the unconditional one on these particular realized lines, since pointwise codeword length is not bounded by entropy (the inequality $H(l_i \mid p) \leq H(l_i)$ governs only *expected* optimal length, not the length of a fixed outcome). When $R \leq 0$ the residual already matches or exceeds the raw cost and the criterion keeps the lines regardless of $L(p)$; when $R > 0$ but $R \leq L(p)$ the savings fail to amortize the summary, and again raw wins. Synthesis is justified exactly at $R > L(p)$, and not before. □

In fabius. archivum folds a grown batch of log lines into a synthesis page only when the page's description cost $L(p)$ is strictly paid back by net cross-line redundancy $R = \sum_i (L(l_i) - L(l_i \mid p)) > L(p)$; until the log accumulates enough genuinely shared structure to make that two-part code shorter, the raw lines stay, so synthesis fires precisely at the description-length break-even and never on a still-incompressible log.

4.6 · Operational heuristics

R4 Scout wide: hold candidate routes until the signal sharpens QUALITATIVE

Rule. On an ambiguous task, fabius keeps two or three candidate routes alive and commits to one only as the signal sharpens — scout wide, then strike narrow. It holds the alternatives as a weighted preference over routes rather than guessing a single route early, and collapses to one once the integrated evidence separates them.

Why it holds. This is an operational heuristic, not a theorem: an early hard commit under ambiguity risks the wrong route with no recovery, while carrying a small set of routes costs little and lets cheap early signal decide. It carries no objective function and never overrides a value-of-information gate — it only defers the collapse until the gate can fire on evidence. It is fabius's own maxim, scout wide / strike narrow, applied to routing.

R10 **State breadth tasks once; hard-steer only narrow contracts** QUALITATIVE

Rule. fabius states a breadth task once and lets the specialist range; it hard-steers only a narrow contract. Instruction emphasis is a dial — turn it up to force adherence to a hard constraint, keep it modest when the breadth of candidate outputs matters — and it measures both adherence and breadth rather than maxing emphasis blindly.

Why it holds. This is an operational heuristic, not a theorem: over-steering a task that wanted range collapses the output to a narrow mode; under-steering a task with a hard constraint lets it drift. The honest, verifiable core is directional — raising emphasis raises adherence and lowers breadth — so fabius tunes emphasis to the task's kind and checks the trade rather than assuming a fixed optimum.

M8b **Break index ties by recency — an order-only tie-break** QUALITATIVE

Rule. When archivum's M8 retrieval leaves a relevance tie on a curated wiki, fabius breaks it toward the more recently edited page — a deterministic recency tie-break, applied only after relevance and never as a scoring stage that can override it.

Why it holds. This is an operational heuristic, not a posterior: on a tie, recency is a cheap, monotone, order-only signal (more recent first) that carries no probabilistic claim and cannot outrank relevance. fabius keeps only the monotone tie-break and deliberately drops any fitted decay rate or importance weighting, so no claim of optimality or calibration is made or needed.

4.7 · The operational edge, proven — R11–R13, M9

These four rules dispatch model tiers, run long-horizon loops, compose verticals, and package the corpus. Each carries a full proof in the house style below — R11 as an index-policy theorem (the tier cascade at cost-per-success, a reservation-value ordering collapsing to R3’s gate verbatim), R12 as optimal stopping with a dual exit gate and an almost-sure termination bound (M4’s objects lifted to cycle scale), R13 as a conflict-freedom / termination / goal-preservation theorem over the single-owner composition, and M9 as a lossless-paging dominance bound with an explicit recall threshold. Every entry was adversarially verified by two independent skeptic passes; the corrections are recorded in the receipt (`paper/proofs.json`), per the house discipline.

R11 The tier cascade is an index policy: dispatch at cost-per-success, escalate on a verified miss

REAL-MATH

Statement. Route a sub-task down a ladder of model tiers ordered by price, escalating only when the R8 verifier certifies a miss. **Claim.** Among all dispatch policies that keep attempting untried tiers until the first verified success — a class in which every policy achieves the identical success guarantee — expected spend is minimized by the static cascade sorted by the reservation index $\rho_i := c_i/p_i$ (price per unit of success probability), and each dispatch decision is one expected-loss threshold of exactly R3's form: attempt tier i ahead of the rest of the plan iff the spend it stands to cancel exceeds its price, $p_iW > c_i$. Cheapest-first is the optimal cascade whenever the price order agrees with the ρ-order — which is what the operational clause "the cheapest tier that holds" enforces — and it is strictly suboptimal whenever a reachable cheap tier's success probability is too thin to earn its slot, which is why ambiguity, architecture, security and irreversible actions route straight to the strong tier by the same index, not by exception.

Setting. Fix a sub-task t. Tiers $i \in \{1, \dots, K\}$ have attempt costs $0 < c_1 \le c_2 \le \cdots \le c_K$ (each c_i includes that attempt's R8 verification cost) and per-sub-task success probabilities $p_i := p_i(t) \in (0, 1]$; attempt outcomes are independent across tiers. The verifier is sound and complete: an attempt passes iff it genuinely clears the sub-task's bar. A *policy* attempts untried tiers one at a time — possibly adaptively, possibly at random — stopping at the first pass; if all K tiers fail it pays a fixed fallback $F \ge 0$ (human escalation, R7/M2). We compare policies in the *equal-guarantee class*: no policy invokes the fallback while an untried tier remains. Every policy in the class succeeds with probability $1 - \prod_{i=1}^{K}(1-p_i)$ — the guarantee is order-invariant, so ordering affects only spend. For a static order (permutation) σ,

$$\mathbb{E}[C(\sigma)] \;=\; \sum_{j=1}^{K} c_{\sigma(j)} \prod_{l=1}^{j-1}\bigl(1-p_{\sigma(l)}\bigr) \;+\; F\prod_{l=1}^{K}\bigl(1-p_{\sigma(l)}\bigr),$$

since tier $\sigma(j)$ is paid iff the first $j-1$ attempts all failed. The F-term is permutation-invariant and drops out of every comparison below.

Lemma 1 (adaptivity collapses to a static order). The minimum of expected spend over all policies in the class equals the minimum over the $K!$ static orders. *Proof.* The only information a policy ever observes is the verifier's verdict, and a pass terminates the cascade. Hence along the unique non-terminated path every verdict is "fail", so a deterministic adaptive policy's successive choices are all made on that one history and trace one fixed permutation: deterministic-adaptive equals static. A randomized policy is a probability mixture over deterministic ones, and its expected spend is the corresponding convex combination of finitely many static-order spends, which is minimized at a pure permutation. □

Lemma 2 (exchange). Let σ be any order and σ' the same order with adjacent slots $j, j+1$ transposed; write $i = \sigma(j)$, $k = \sigma(j+1)$, and $q = \prod_{l<j}\bigl(1-p_{\sigma(l)}\bigr)$ for the probability of reaching slot j. Then

$$\mathbb{E}[C(\sigma)] - \mathbb{E}[C(\sigma')] \;=\; q\,\bigl(p_k c_i - p_i c_k\bigr) \;=\; q\,p_i p_k\,(\rho_i - \rho_k),$$

so placing i before k is weakly better if $\rho_i \le \rho_k$ (an equivalence when $q > 0$; at $q = 0$ the two orders cost the same), and strictly better iff $\rho_i < \rho_k$ and $q > 0$. *Proof.* Slots before j contribute identically. The pair contributes $q\,[c_i + (1-p_i)c_k]$ under σ versus $q\,[c_k + (1-p_k)c_i]$ under σ'. Slots after $j+1$ contribute identically: they are reached with probability $q(1-p_i)(1-p_k)$ in either order — symmetric in i, k — with the same set of tried tiers, hence identical continuations. Subtracting the pair terms gives $q(p_k c_i - p_i c_k)$; dividing by $p_i p_k > 0$ shows its sign is that of $\rho_i - \rho_k$. □

Theorem (index-cascade optimality). (a) Any static order nondecreasing in ρ minimizes expected spend over the entire equal-guarantee class, and all ρ-nondecreasing orders have equal spend. (b) In particular the cheapest-first cascade $1, 2, \dots, K$ is optimal whenever $\rho_1 \le \rho_2 \le \cdots \le \rho_K$; conversely, if $\rho_j > \rho_{j+1}$ for some j and slot j is reached with positive probability ($p_l < 1$ for all $l < j$), cheapest-first is strictly suboptimal — the exchange strictly saves.

Proof. (a) Take any permutation σ. While some adjacent pair satisfies $\rho_{\sigma(j)} > \rho_{\sigma(j+1)}$, transpose it: by Lemma 2 the expected spend weakly decreases, and the transposition changes the relative order of that pair only, so the count of strict ρ-inversions falls by exactly one. The procedure therefore terminates, at an order with no adjacent strict inversion — i.e. a ρ-nondecreasing order — whose spend is at most $\mathbb{E}[C(\sigma)]$. Hence some sorted order attains the minimum over permutations, and by Lemma 1 over the whole class. Any two ρ-nondecreasing orders can differ only in the arrangement of equal-ρ tiers, which occupy the same consecutive block of slots in both (the block is fixed by the counts of strictly smaller and equal index values); within a block any rearrangement is a product of adjacent transpositions of equal-ρ pairs, each changing spend by $q\,p_i p_k(\rho_i - \rho_k) = 0$ by Lemma 2. So all sorted orders cost the same. (b) The first claim is a case of (a). For the converse, apply Lemma 2 to the identity order at slots $j, j+1$: the swap changes spend by $q p_j p_{j+1}(\rho_j - \rho_{j+1}) > 0$ since $q > 0$, so the identity is not optimal. □

Threshold form (the gate the coherence extension composes). The theorem compresses to one scalar gate. Let W be the expected spend of any plan for the sub-task that does not attempt tier i (the rest of the cascade, or the strong tier directly). Prepending a verified attempt of tier i costs c_i and with probability p_i cancels W entirely, so the plan-with-probe spends $c_i + (1-p_i)W$. Prepend iff

$$c_i + (1-p_i)\,W \;<\; W \quad\Longleftrightarrow\quad p_i W > c_i \quad\Longleftrightarrow\quad W > \rho_i.$$

This is R3's gate verbatim, on disjoint variables: $a = W$ (expected spend acting without the probe), $b = (1-p_i)W$ (expected residual spend after it), $\mathrm{EVOI} = a - b = p_i W$, $c_{\mathrm{call}} = c_i$, and the gate fires iff $\mathrm{EVOI} > c_{\mathrm{call}}$. Equivalently ρ_i is tier i's *reservation value* — the downstream-spend level at which attempting it exactly breaks even — and the ρ-sorted cascade is the greedy reservation-value policy (equivalently, the reservation-value ordering; the c/p exchange index is the sequential-testing index). The gate prices *where* a tier goes, not *whether* it runs: participation is pinned by the equal-guarantee class, and letting the same gate also truncate the cascade — skip tier i straight to the fallback when $W \le \rho_i$ — changes the success guarantee and exits the class, so no optimality is claimed for truncation here. Anchor case $K = 2$, $p_2 = 1$: try the cheap tier first iff $p_1 > c_1/c_2$ — at a tenfold price gap the cheap tier earns its slot iff its verified success probability exceeds 10%.

Corollaries — one per clause of the shipped rule. (i) *Re-tier per sub-task.* The index is $\rho_i(t) = c_i/p_i(t)$: prices are global but success probabilities are per-sub-task. If sub-tasks t, t' order two tiers oppositely ($\rho_i(t) < \rho_k(t)$ but $\rho_i(t') > \rho_k(t')$), any single session-wide order inverts the pair $\{i, k\}$ for one of them, hence is not ρ-sorted for that sub-task; a non-sorted order has an adjacent strict inversion, and when that slot is reachable the exchange of Lemma 2 strictly saves — the session-wide order strictly overspends there. (ii) *Reserve the strong tier — by index, not exception.* On ambiguity, architecture and security sub-tasks $p_{\mathrm{cheap}}(t)$ is small, so $\rho_{\mathrm{cheap}} = c_{\mathrm{cheap}}/p_{\mathrm{cheap}}$ blows up past ρ_{strong} and the sorted cascade opens with the strong tier: dispatching straight to strength on hard calls is the same index policy. For irreversible actions, let a tier-i miss additionally cause expected damage d_i before the verifier catches it; every occurrence of c_i in the spend formula and in Lemma 2 becomes the expected outlay $\tilde{c}_i = c_i + (1-p_i)d_i$, the entire argument runs unchanged, and the inflated index $\tilde{\rho}_i = \tilde{c}_i/p_i$ pushes damage-prone cheap tiers behind the strong one. (iii) *Escalate on a verified miss, never a hunch.* Modify the optimal cascade so that at a slot j with nonempty continuation the agent, with probability $\beta > 0$, escalates without consulting the verifier. With probability $q_j\,\beta\,p_j$ a genuine success is discarded and the continuation spend $W_{j+1} > 0$ (positive since all $c_i > 0$) is paid needlessly: the expected excess $q_j\,\beta\,p_j\,W_{j+1} > 0$ whenever slot j is reachable, and the discarded success is recovered only if a later tier passes, so the guarantee cannot rise — hunch-escalation is dominated on both coordinates. Dually, *accepting* on a hunch at the same slot j — stopping unverified with probability $\alpha > 0$, blind to the attempt's true outcome — costs nothing when the attempt happened to succeed but strands the sub-task when it failed: it lowers the success probability by $q_j\,\alpha\,(1-p_j)\,S_{j+1}$, where S_{j+1} is the continuation's success probability — positive here, since the continuation is nonempty and every $p_i > 0$ — and so exits the equal-guarantee class altogether whenever slot j is reachable. (At the final slot the continuation is empty and $S_{K+1} = 0$; there a blind accept instead swallows the fallback F, reporting an unverified outcome as done — it violates the class's stopping rule directly rather than through the guarantee.) Hence verified-miss escalation is the only admissible trigger in the class, and among admissible policies the Theorem selects the index cascade. □

In fabius. R11 dispatches each sub-task to the ρ-minimal tier and lets R8's verdict — never a vibe — trigger the climb. "The cheapest tier that holds" is exactly the condition that the price order and the ρ-order coincide: mechanical, tightly-contracted work has $p_{\mathrm{cheap}} \approx 1$, so $\rho_{\mathrm{cheap}} \approx c_{\mathrm{cheap}}$ and stays cheap, while ambiguity, security calls and irreversibility inflate ρ_{cheap} (through small p or damage d) and route to strength via the same single threshold $W > \rho_i$ that R3 applies to calls and R2 applies to ladder rungs.

Boundary. PROVED, under the stated model: Lemmas 1–2, the Theorem, the threshold/EVOI form, and corollaries (i)–(iii) — assuming (A1) attempt outcomes independent across tiers, (A2) a sound and complete verifier with its cost folded into c_i , (A3) known, calibrated $(c_i, p_i(t))$, (A4) risk-neutral expected spend with one attempt per tier. What remains operational heuristic: independence (A1) is load-bearing — real tier failures correlate through latent sub-task difficulty, a cheap-tier miss is evidence lowering every untried tier's posterior success probability, and under correlation the static index policy loses its optimality proof (recomputing indices on posterior probabilities is the sensible move, but no optimality is claimed here, and optimal correlated sequencing is intractable in general); same-tier retries are the maximally correlated case and sit outside the model. Corollary (iii) prices a hunch-escalated attempt at the full c_j — verification folded in per (A2) — so a policy that skips verification precisely to recoup its cost sits outside the model and is not dominated by this argument; it is excluded on other grounds (an unverified pass certifies nothing). R8's real verifier is only approximately sound and complete — a false pass erodes the "same success guarantee" clause at a rate this entry does not bound. The indices themselves are estimates: the theorem is conditional on calibration, and no robustness bound for misordered ρ's is proved. Which sub-tasks count as "ambiguity / architecture / security" is a classification judgment about where p_{cheap} is small, not a theorem. Finally, headline savings figures are task- and dataset-specific; what transfers — and what is proved — is the index/threshold form, not any particular cost number.

R12 Dual exit gate + hard cap: the long-horizon step→verify loop as optimal stopping

REAL-MATH

Statement. For multi-cycle autonomous work, run the `fabius-disciplina step → verify` cycle as a stochastic loop with exactly three exits: *stop-done* at the first cycle where the completion condition C_k and an explicit, separately-sourced done-signal D_k hold simultaneously; *stop-escalate* on a certified stall; and a hard cap of N cycles, rate-limited and checkpointed. We prove three things. (i) The conjunctive gate $C_k \wedge D_k$ never has a higher false-stop rate than either signal alone, shrinks it *multiplicatively* when the two signals are conditionally independent — the design target that *separately sourced* operationalizes — and pays at worst an *additive* price in missed stops; over an N-cycle horizon exactly this shrinkage is what keeps the run-level false-stop probability bounded. (ii) The loop terminates surely with deterministically bounded cost and bounded state corruption, and both stall tests are inherited objects: the no-progress window is a hypothesis test whose null is R8's contraction and whose size is at most L^w, and the repeated-cause reflection test is M4's redundancy lemma verbatim. (iii) The whole rule is one expected-loss threshold — continue iff $(1-q_k)\,p_{k+1} > c_{\text{cyc}}/W$, M4's threshold functional applied to the disjoint variable pair (completion belief, per-cycle hazard) — and with the hard cap making the horizon finite, under a monotone (diminishing-returns) hazard this myopic threshold is the *exact* optimal stopping rule for the shipped declare-done loss, not merely a heuristic.

Setting. Cycles $k = 1, 2, \ldots$ are adapted to a filtration $(\mathcal{F}_k)$. Let T_k be the event that the task is truly complete after cycle k; completion is absorbing, $T_k \subseteq T_{k+1}$. Write $q_k = \mathbb{P}(T_k \mid \mathcal{F}_k)$ (completion belief) and $p_{k+1} = \mathbb{P}(T_{k+1} \mid \neg T_k,\ \mathcal{F}_k)$ (per-cycle hazard); on $\{q_k = 1\}$ the hazard is vacuous and we adopt the convention $(1-q_k)p_{k+1} = 0$. Each cycle emits two stop signals: C_k, the verify condition read from the objective verifier, and D_k, an explicit done-signal from a distinct channel (sentinel marker, independent checker, human ack). Error rates, taken uniform across cycles: false alarms $\alpha_C = \mathbb{P}(C_k \mid \neg T_k)$, $\alpha_D = \mathbb{P}(D_k \mid \neg T_k)$; misses $\beta_C = \mathbb{P}(\neg C_k \mid T_k)$, $\beta_D = \mathbb{P}(\neg D_k \mid T_k)$. Costs: per-cycle cost $c_{\text{cyc}} \in [c_{\min}, c_{\max}]$ with $c_{\max}$ enforced by the rate limit; $W > 0$ the loss of a false stop (handing off incomplete work as done); $c_{\text{esc}} \in (0, W)$ the cost of escalating. From R8 reuse the progress metric $e_k = d(x_k, x^*) \ge 0$, the distance of the working state to the verify condition's fixed point.

Theorem. (i) *Gate.* Unconditionally $\mathbb{P}(C_k \wedge D_k \mid \neg T_k) \le \min(\alpha_C, \alpha_D)$. If $C_k \perp D_k \mid \neg T_k$, it equals $\alpha_C\alpha_D$, strictly below the min whenever both rates lie in $(0,1)$; meanwhile $\mathbb{P}(\neg(C_k \wedge D_k) \mid T_k) \le \beta_C + \beta_D$. Over a run of at most N gate evaluations the run-level false-stop probability is at most $N\alpha_C\alpha_D$ (dual, independent) versus $N\alpha_C$ (single). (ii) *Termination.* $\tau = \min(\tau_{\text{gate}}, \tau_{\text{stall}}, N) \le N$ surely; total cost $\le N c_{\max}$; a runaway rolls back at most r cycles of state, r the checkpoint interval. If the loop still contracts in R8's sense ($\mathbb{E}[e_{k+1} \mid \mathcal{F}_k] \le L\, e_k$, $L < 1$), a w-cycle no-progress window fires falsely with probability at most L^w. (iii) *Threshold and optimality.* One-step lookahead yields the policy: continue iff $(1-q_k)p_{k+1} > c_{\text{cyc}}/W$, with ties broken toward stopping — at equality the two actions tie exactly, and stopping is the safe tie-break; at stop, declare done iff $q_k > 1 - c_{\text{esc}}/W$, else escalate. With horizon capped at N, constant per-cycle cost, and the monotone hypothesis (once the one-step continue value is non-positive it stays non-positive a.s.), the first-crossing rule is exactly optimal among all stopping times for the declare-done terminal loss, and every optimal rule satisfies $\mathbb{E}[\sigma^*] \le W/c_{\min}$.

Proof. *(i) False stops multiply down; misses add up.* $\{C \wedge D\} \subseteq \{C\}$ and $\subseteq \{D\}$, so monotonicity of measure gives $\mathbb{P}(C \wedge D \mid \neg T) \le \min(\alpha_C, \alpha_D)$ with no assumptions — the dual gate is never worse. Under conditional independence given $\neg T$ the conjunction factorizes, $\mathbb{P}(C \wedge D \mid \neg T) = \alpha_C\alpha_D$, and $\alpha_C\alpha_D < \min(\alpha_C, \alpha_D)$ iff $\min > 0$ and $\max < 1$ — a strict multiplicative drop. On the miss side, $\mathbb{P}(\neg(C \wedge D) \mid T) = \mathbb{P}(\neg C \cup \neg D \mid T) \le \beta_C + \beta_D$ by the union bound: the price of demanding both is at worst additive in the individual miss rates while the benefit is multiplicative in the false-alarm rates — an asymmetric trade that favors the conjunction precisely when the signals are individually reliable on true completion (small β) but individually gameable on non-completion (moderate α), which is the LLM self-assessment regime. Evidence form: if additionally $C \perp D \mid T$, then $\operatorname{odds}(\neg T{:}T \mid C \wedge D) = \dfrac{\alpha_C\,\alpha_D}{(1-\beta_C)(1-\beta_D)} \cdot \operatorname{odds}(\neg T{:}T)$ — the log-Bayes-factors add, so a bar on the posterior that neither channel clears alone can be cleared conjunctively. Horizon necessity: a false stop at cycle k implies $G_k \wedge \neg T_k$ for the gate event G_k, so $\mathbb{P}(\text{any false stop}) \le \sum_{k \le N} \mathbb{P}(G_k \wedge \neg T_k) \le N \max_k \mathbb{P}(G_k \mid \neg T_k)$: certifying a run-level budget ε through this bound requires a per-cycle rate at most ε/N, which shrinks with the horizon — the dual gate's product is what supplies that headroom without requiring a better single verifier. Dependence, honestly: in general $\mathbb{P}(C \wedge D \mid \neg T) = \kappa\,\alpha_C\alpha_D$ for some κ in the Fréchet range $\max(0,\ \alpha_C + \alpha_D - 1)/(\alpha_C\alpha_D) \le \kappa \le 1/\max(\alpha_C, \alpha_D)$ (upper end by the min bound; the lower end, $\kappa < 1$, is negative dependence and only helps). The failure mode is positive dependence, $\kappa > 1$: κ climbs toward its maximum exactly as the two signals collapse into one source — e.g. both computed from the model's own self-assessment — and at the top of the range the gate degrades to the better single signal, never below it. This is the mathematical content of requiring D to be *separately sourced*: separation is what drives κ down to 1 and turns the min bound into the product.

(ii) Sure termination, bounded blast radius, and inherited stall tests. $\tau \le N$ holds deterministically, which is stronger than almost-sure termination; total cost $\sum_{k\le\tau} c_k \le N c_{\max}$ since the rate limit enforces $c_k \le c_{\max}$; and with a checkpoint every r cycles, a detected runaway rolls back to a checkpoint at distance at most r, bounding corruption by r cycles of mutations. These three are properties of the *enforced harness*, not of the agent's judgment — which is the point: they hold even when q and p are estimated badly. Stall test: under the contraction null, iterating the tower property gives $\mathbb{E}[e_{k+w} \mid \mathcal{F}_k] \le L^w e_k$, and a tail (concentration) bound ($e \ge 0$; $e_k > 0$, else the verify condition already holds) yields $\mathbb{P}(e_{k+w} \ge e_k \mid \mathcal{F}_k) \le L^w$. At R8's moderate default $L \approx \frac{1}{2}$ with $w = 3$, a still-contracting loop shows a flat window with probability at most $1/8$, while R8's no-guarantee regimes admit no such bound: the non-expansive isometry ($e_{k+1} = e_k$) fires the window with probability 1, and the martingale regime ($\mathbb{E}[e_{k+1} \mid \mathcal{F}_k] = e_k$) contains the isometry as a special case, so no bound below 1 exists there. The window is therefore a proper hypothesis test: size at most L^w under the contraction null, decaying geometrically in w, with power 1 against the stalled-at-a-fixed-distance alternative. Reflection test: by M4's lemma, $I(R_k; \mathrm{cause} \mid R_{<k}) = 0$ leaves the posterior over the cause unchanged; the next attempt, drawn by the same policy from that unchanged posterior, has the same success law, hence $p_{k+1} = p_k$ — a reflection that repeats the prior cause with no new hypothesis cannot lift the hazard over a threshold it already failed, so the next cycle is the same losing bet: stop and escalate. Finally, in the good regime the cap is slack: contraction gives $\mathbb{E}[e_k] \le L^k e_0$, so the expectation falls below tol within $\lceil \ln(e_0/\mathrm{tol})/\ln(1/L) \rceil$ cycles (R8's cap computation) and past that point a tail bound controls $\mathbb{P}(e_k > \mathrm{tol})$ by a geometrically decaying tail; a cap at that scale binds only on runs where the contraction hypothesis has already failed with high probability — the cap is insurance, not the mechanism.

(iii) One threshold; exact optimality under the cap. Absorption gives $\mathbb{P}(\neg T_{k+1} \mid \mathcal{F}_k) = \mathbb{P}(\neg T_k \mid \mathcal{F}_k)\,\mathbb{P}(\neg T_{k+1} \mid \neg T_k, \mathcal{F}_k) = (1-q_k)(1-p_{k+1})$, and by the tower property $\mathbb{E}[1-q_{k+1} \mid \mathcal{F}_k]$ equals the same quantity. Stopping now and declaring done loses $(1-q_k)W$ in expectation; one more cycle and then stopping loses $c_{\mathrm{cyc}} + (1-q_k)(1-p_{k+1})W$. Continuing dominates iff $c_{\mathrm{cyc}} < (1-q_k)p_{k+1}W$, i.e.

$$(1-q_k)\,p_{k+1} \;>\; \frac{c_{\mathrm{cyc}}}{W},$$

which is M4's threshold $p > c_{\mathrm{retry}}/V_{\mathrm{fix}}$ with the belief factor exposed: after a fresh observed failure M4 has $q_k = 0$ and the two coincide; at equality the two actions tie and the rule stops. At stop, declaring done loses $(1-q_k)W$ and escalating loses c_{esc}, so declare done iff $q_k > 1 - c_{\mathrm{esc}}/W$ — a second expected-loss threshold, on q_k alone, and the dual gate is its implementation: by (i)'s additive log-evidence, $C_k \wedge D_k$ pushes q_k over the bar exactly when the combined Bayes factors suffice. (Folding escalation into the lookahead replaces the terminal loss by $\min((1-q)W,\ c_{\mathrm{esc}})$. Because $\min$ is concave the exact continue test becomes a value-of-information inequality, $\mathbb{E}[\min((1-q_{k+1})W, c_{\mathrm{esc}}) \mid \mathcal{F}_k] < \min((1-q_k)W, c_{\mathrm{esc}}) - c_{\mathrm{cyc}}$; under a posterior-resolving gate — the next gate reading collapses q_{k+1} into $\{0,1\}$ — it evaluates in the escalate regime to $q_k + (1-q_k)p_{k+1} > c_{\mathrm{cyc}}/c_{\mathrm{esc}}$: the same threshold form with two changes, both exact for this loss. The bar rises from c_{cyc}/W to $c_{\mathrm{cyc}}/c_{\mathrm{esc}}$ — strictly larger since $c_{\mathrm{esc}} < W$: with escalation in hand only the recoverable stake rides on the outcome, so a cycle must earn more to be worth buying. And the left side gains the resolution term q_k — what a cycle now buys is the chance the gate *certifies* completion, an event of probability $q_k + (1-q_k)p_{k+1}$ — so the folded rule is not uniformly stricter: it can rightly continue where the plain rule stops, when q_k is substantial but under the declare bar and one cycle suffices to resolve it. The form composes; only this one-step test, not first-crossing optimality, is claimed for the folded loss.) The two stop modes are the two collapses of the continue value $(1-q_k)p_{k+1}$: $q_k \to 1$ (gate fired: stop-done) or p_{k+1} stalled with q_k low (stop-escalate). "Never spin" is the statement that no third mode exists: below threshold, continuation is dominated.

Myopic $\Rightarrow$ optimal. The cap makes the horizon finite, so the stopping problem is well posed with bounded losses. Let $\ell_j = (1-q_j)W$ and take the per-cycle cost constant at c_{cyc} (an $\mathcal{F}_j$-measurable varying cost c_j only re-indexes the stop region by c_j/W; the identical computation goes through). Define the one-step region $S_j = \{(1-q_j)p_{j+1} \le c_{\mathrm{cyc}}/W\} \in \mathcal{F}_j$. For any stopping time σ with $k \le \sigma \le N$, telescope the loss: $\ell_\sigma + (\sigma - k)c_{\mathrm{cyc}} = \ell_k + \sum_{j=k}^{\sigma-1} Y_j$ with $Y_j = \ell_{j+1} - \ell_j + c_{\mathrm{cyc}}$ and $\mathbb{E}[Y_j \mid \mathcal{F}_j] = c_{\mathrm{cyc}} - (1-q_j)p_{j+1}W \ge 0$ on S_j. Under the monotone hypothesis — entering S means staying, $S_k \subseteq S_j$ a.s. for $j \ge k$; the diminishing-returns regime, of which M4's geometric hazard decay is a special case — every increment after entry is conditionally nonnegative, and since $\{\sigma > j\} \in \mathcal{F}_j$, $\mathbb{E}\big[\sum_j \mathbf{1}\{\sigma > j\} Y_j \,\big|\, \mathcal{F}_k\big] = \sum_j \mathbb{E}\big[\mathbf{1}\{\sigma > j\}\,\mathbb{E}[Y_j \mid \mathcal{F}_j] \,\big|\, \mathcal{F}_k\big] \ge 0$ on S_k. Hence every σ does at least as badly as stopping at once on S_k; off S_k the inequality $(1-q_k)p_{k+1} > c_{\mathrm{cyc}}/W$ is strict, so taking one more cycle strictly beats stopping: no optimal rule stops off S before the cap (at $j = N$ the harness forces the stop wherever the process sits). First crossing of the threshold — or the cap, if the threshold is never crossed — is therefore an exact optimum: stopping off S before the cap is strictly dominated, and on S no continuation improves on stopping.

The cap's scale is principled. Any rule σ pays at least $\sigma c_{\min}$, and the optimum's expected loss is at most that of stopping immediately, itself at most W; hence $\mathbb{E}[\sigma^*]\, c_{\min} \le W$, so $\mathbb{E}[\sigma^*] \le W/c_{\min}$ and $\mathbb{P}(\sigma^* \ge m\, W/c_{\min}) \le 1/m$. The rational cycle budget is intrinsically bounded by the ratio of ship-incomplete loss to minimum cycle cost; a hard cap at scale $N \approx W/c_{\min}$ removes only behavior the optimum already excludes in expectation, while protecting against the risk the model cannot see from inside: a miscalibrated $\hat{q}$ or $\hat{p}$. □

In fabius. R12 wraps long-horizon autonomous work in the `step → verify` loop and exits only through the threshold: keep cycling while $(1-q_k)p_{k+1} > c_{\text{cyc}}/W$; stop-done when the dual gate $C_k \wedge D_k$ — objective verify condition *plus* separately-sourced done-signal — pushes q_k over $1 - c_{\text{esc}}/W$; stop-escalate when a flat $w \approx 3$-cycle progress window (R8's contraction refuted at test size L^w) or a cause-repeating reflection (M4's zero-information lemma) certifies the hazard has stalled below the bar; and the harness, not the model, enforces the cap N, the rate limit $c_{\max}$, and checkpoints every r cycles.

Boundary. PROVED under stated hypotheses: the min bound, the additive miss bound, and the Fréchet range for κ (assumption-free); the multiplicative false-stop drop and additive log-evidence (conditional independence of C, D given the truth); sure termination, cost $\le Nc_{\max}$, and corruption $\le r$ cycles (given the harness enforces cap, rate limit, and checkpoints *externally* to the looping model); the stall test's size $\le L^w$ (under the stochastic-contraction null inherited from R8); the redundancy lemma (M4, verbatim, plus the stated same-policy premise); the threshold form (one-step lookahead); and exact optimality of the first-crossing-or-cap rule plus $\mathbb{E}[\sigma^*] \le W/c_{\min}$ (finite horizon + monotone hazard + constant per-cycle cost, for the declare-done terminal loss). OPERATIONAL, not proved: independence of C and D is a design target, not a theorem — the dependence factor κ must be measured per deployment, and under full positive dependence the gate degrades, safely, to the better single signal; the stall test's L^w is a size bound only — its power is 1 against the stalled-at-fixed-distance alternative but unquantified against slow leaks hovering just inside the null; folding the escalate option in is established at one-step lookahead only — closed-form under a posterior-resolving gate, an open value-of-information inequality otherwise — and first-crossing optimality is proved for the declare-done loss alone, not the folded loss; the specific constants — window $w \approx 3$, the cap value, the checkpoint interval r — are heuristics for which the theory fixes only scales (w via the test size L^w, N via $W/c_{\min}$, r via tolerable rework), not values; q_k and p_{k+1} are estimated by the very model that is looping, and a self-assessment-biased $\hat{q}$ is precisely the failure mode the separately-sourced D and the external cap exist to bound; and where the monotone hypothesis fails — fresh information genuinely revives the hazard — the first-crossing rule stops early and escalates when an oracle might have continued: the safe side of wrong, but not the optimal one.

R13 A vertical runs a studio — domain-led composition is conflict-free, terminating, and goal-preserving

REAL-MATH

Statement. A vertical task (a game, a launch, a security review, a landing page) spans several concern-blocks at once. Model its concerns as a finite variable set V and lift R1's single-owner structure from tasks to concerns: an ownership map $\omega : V \to \Lambda$, $\Lambda = \{D, P, E, F\}$ (domain leader, disciplina, decor·cohors, parcus), assigns each concern exactly one owner. We prove that if every layer constrains only the block it owns, the studio composition $D \prec P \prec E \prec F$ is (i) well-defined and conflict-free, (ii) acyclic and terminating in one pass, and (iii) exactly goal-preserving — and, conversely, that collapsing a multi-block vertical to a single layer leaves every other loaded block uncertified, while a downstream locality violation can both redefine the domain goal and empty the pipeline. The routing gate is one threshold on one fresh variable: the loaded-block count $k(t)$, with studio iff $k(t) \geq 2$.

Setting. Each concern $v \in V$ has a nonempty value domain X_v; the outcome space is $\Omega = \prod_{v \in V} X_v$. By R1(i) applied to ω — preimages of disjoint singletons — the blocks $V_\lambda = \omega^{-1}(\{\lambda\})$ are pairwise disjoint and exhaust V: a measurable partition of concern-space. Write $\Omega_\lambda = \prod_{v \in V_\lambda} X_v$ (a one-point space if $V_\lambda = \varnothing$, by the empty-product convention) and $\pi_\lambda : \Omega \to \Omega_\lambda$ for coordinate projection. *Locality axiom* (the single-owner discipline): layer λ acts by publishing a constraint only on what it owns — a cylinder $K_\lambda = \pi_\lambda^{-1}(A_\lambda)$ with $\varnothing \neq A_\lambda \subseteq \Omega_\lambda$ — and operating on candidate sets $S \subseteq \Omega$ as $T_\lambda(S) = S \cap K_\lambda$. The domain leader's constraint carries the goal: $g := K_D$. Call λ *loaded* iff $A_\lambda \subsetneq \Omega_\lambda$ (its constraint genuinely excludes something), and let $k(t) = |\{\lambda : \lambda \text{ loaded}\}|$.

Theorem (studio composition). Under single ownership and locality: (i) the composed output $S^* = (T_F \circ T_E \circ T_P \circ T_D)(\Omega)$ equals $\bigcap_\lambda K_\lambda$, is independent of firing order, factorizes as $S^* \cong \prod_{\lambda \in \Lambda} A_\lambda$, and is nonempty — in particular no two layers can publish contradictory constraints and no concern is legislated twice; (ii) the pipeline is a finite strict order, hence acyclic, and one topological pass reaches S^*, a fixed point of every T_λ; (iii) $S^* \subseteq g$ and $\pi_D(S^*) = A_D$: downstream layers refine strictly within their own coordinates and neither violate nor reshape the WHAT — and locality is necessary, since its failure admits both goal-redefinition and vacuity; (iv) (*collapse lemma*) if $k(t) \geq 2$ and the task is dispatched to a single layer λ, then for every other loaded μ the dispatch admits an outcome violating K_μ; and when the domain block is itself loaded ($A_D \subsetneq \Omega_D$ — a goal that excludes something, the defining mark of a vertical), collapsing to any downstream layer admits outcomes violating g itself.

Proof. *(i) Well-defined, conflict-free.* Because $\{V_\lambda\}_{\lambda \in \Lambda}$ partitions V (totality of ω gives exhaustion, single-valuedness gives disjointness — R1's partition doing the structural work), coordinate evaluation $x \mapsto (\pi_\lambda(x))_{\lambda \in \Lambda}$ is a bijection $\Omega \to \prod_\lambda \Omega_\lambda$: every coordinate appears in exactly one factor. For $x \in \Omega$: $x \in \bigcap_\lambda K_\lambda \iff \pi_\lambda(x) \in A_\lambda \ \forall \lambda \iff (\pi_\lambda(x))_\lambda \in \prod_\lambda A_\lambda$, hence

$$\bigcap_{\lambda \in \Lambda} K_\lambda \ \cong \ \prod_{\lambda \in \Lambda} A_\lambda.$$

Intersection is commutative and associative, so every firing order yields the same $S^* = \bigcap_\lambda K_\lambda$. Each $A_\lambda \neq \varnothing$ and Λ is finite, so choosing one $a_\lambda \in A_\lambda$ per layer (finite choice, no axiom needed) exhibits a point of S^*: $S^* \neq \varnothing$. The same construction for any pair $\lambda \neq \mu$ shows $K_\lambda \cap K_\mu \neq \varnothing$: cylinders over disjoint blocks cannot contradict each other, so an inter-owner conflict is structurally impossible, and each concern v is constrained exactly once — through $A_{\omega(v)}$, by its unique owner.

(ii) Acyclic, terminating. $(\Lambda, \prec)$ with $D \prec P \prec E \prec F$ is a strict total order on a finite set — an acyclic digraph with no back-edge — so the topological pass is its unique linear extension and fires each T_λ exactly once: the pipeline halts after $|\Lambda| = 4$ steps. Each T_λ is intensive ($T_\lambda(S) = S \cap K_\lambda \subseteq S$) and idempotent ($(S \cap K_\lambda) \cap K_\lambda = S \cap K_\lambda$), so $T_\lambda(S^*) = S^*$ for every λ: re-firing any layer is a no-op, the descent in the complete lattice $(2^\Omega, \subseteq)$ cannot oscillate, and the one-pass output is already the fixed point of the whole system.

(iii) Objective preservation, and necessity of locality. $S^* \subseteq K_D = g$, so every composed outcome satisfies the domain goal. For the projection: by (i), $S^* \cong \prod_\lambda A_\lambda$ with every factor nonempty, and the projection of a product of nonempty sets onto one factor is that factor, so $\pi_D(S^*) = A_D$ — downstream layers shrink nothing on V_D: refinement, never redefinition. Necessity: let $V = \{d, e\}$, $X_d = X_e = \{0, 1\}$, $V_D = \{d\}$, $V_E = \{e\}$ (so $V_P = V_F = \varnothing$: A_P, A_F are the one-point spaces, $K_P = K_F = \Omega$, and both drop out of the intersection), and let E break locality by publishing $K_E = \{x : x_d = 1\}$, a constraint on the domain's coordinate. If $A_D = \{0, 1\}$, then $S^* = K_E$ and $\pi_D(S^*) = \{1\} \subsetneq A_D$: the execution layer has silently redefined the WHAT. If $A_D = \{0\}$, then $S^* = K_D \cap K_E = \varnothing$: the pipeline contradicts itself. So dropping locality falsifies conclusion (iii) already at two binary variables — both shipped prohibitions ("don't collapse a vertical", "don't let a downstream layer redefine the goal") are load-bearing axioms, not style.

(iv) Collapse lemma. Suppose $k(t) \geq 2$ and dispatch goes to the single layer λ; the dispatch certifies only membership in K_λ. Let $\mu \neq \lambda$ be any other loaded layer: $A_\mu \subsetneq \Omega_\mu$ yields some $b \in \Omega_\mu \setminus A_\mu$. Choose $a \in A_\lambda$ and, for every remaining block ν, any $c_\nu \in \Omega_\nu$ (all nonempty). The outcome x with $\pi_\lambda(x) = a$, $\pi_\mu(x) = b$, $\pi_\nu(x) = c_\nu$ lies in $K_\lambda \setminus K_\mu$: admitted by the collapsed dispatch, violating μ's constraint. Hence a single-layer collapse leaves every other loaded block uncertified — at least $k - 1 \geq 1$ of them, and all k if λ is itself unloaded. If moreover D is loaded, taking $\mu = D$ shows that collapsing to any downstream layer admits outcomes that violate the goal g itself; if D is unloaded, then $g = \Omega$ is unviolable but also contentless — there is no WHAT, and the task was never a vertical. ☐

In fabius (threshold form). R13's gate variable is the loaded-block count $k(t) \in \{0, \ldots, 4\}$, computed from R1's load bits extended by a domain bit ℓ_D — a coordinate no other rule reads (R2 reads the rung, R3 the tool margin, R7 the width, M6–M8 the store). The gate is the single threshold

$$\textbf{run the studio} \iff k(t) \;\geq\; 2,$$

which implements the house expected-loss inequality $\mathbb{E}[L \mid \textbf{collapse}] - \mathbb{E}[L \mid \textbf{studio}] \;>\; c_{\text{pipe}}$: the studio output S^* satisfies every K_μ by (i), so its constraint-miss loss is zero, while by the collapse lemma the collapse side carries the expected miss-cost $\sum_{\mu \neq \lambda,\, \mu \text{ loaded}} (1 - h_\mu) L_\mu$ of the uncertified blocks (h_μ the chance an unconstrained block lands right anyway, L_μ its miss cost) — positive whenever $k \geq 2$ and some uncertified block has $h_\mu < 1$ and $L_\mu > 0$, the generic case for a genuinely loaded block. At the tie $k \leq 1$ the rule takes the cheaper action — single dispatch, with parcus inline at $k = 0$ — matching R2/R3's cheap-at-tie convention, so R13 composes with the capstone's disjoint-variable argument as one more sign-consistent inequality in one more fresh variable. Inside each studio stage the downstream gates fire unchanged: R2 admits the rung, R6 $\prec$ R5 order execution, decor/cohors execute, parcus floors every stage.

Boundary. Proved: (i)–(iv) are theorems of the model — single ownership plus locality imply conflict-freeness, one-pass termination with a fixed point, exact goal preservation $\pi_D(S^*) = A_D$, the impossibility of inter-owner contradiction, that any single-layer collapse admits outcomes violating every other loaded block's constraint (and the goal itself when the domain block is loaded), and (via the two-variable witnesses) the necessity of the locality axiom. Not proved, and stated as such: (a) *who leads* — the theorem takes "the goal lives in V_D" as given; that the domain layer, rather than process or execution, authors g is fabius's design choice, and since the T_λ commute, the order $D \prec P \prec E \prec F$ matters operationally (who writes before whom reads) rather than set-theoretically; (b) *locality compliance* — that a real LLM layer constrains only its own block, and satisfiably so ($A_\lambda \neq \varnothing$), is a prompt-level contract enforced operationally, not a provable property of the layer; the sharpness examples show exactly what breaks when it slips; (c) *the cap and the costs* — the threshold value $k \geq 2$ and the coordination cost c_{pipe} are operational: the collapse lemma is set-theoretic — it proves violating outcomes are *admitted* — so the expected-loss gap is positive only under the mild extra assumptions $h_\mu < 1$, $L_\mu > 0$ on some uncertified block; the gap's magnitude $\sum (1 - h_\mu) L_\mu$ versus c_{pipe} is estimated in-context per task, not derived, and $k(t)$ itself depends on the router's judgment of which blocks are genuinely loaded; (d) the model abstracts a layer's generative work to its constraint footprint K_λ — content production is richer than the set it certifies.

M9 Externalization is lossless paging that wins once index recall clears an explicit threshold REAL-MATH

Statement. Let a skill's reference corpus be pages $C = \{p_1, \ldots, p_n\}$, $n \geq 1$, with token lengths $\ell(p_j) > 0$ and total $B = \sum_j \ell(p_j)$. Packaging decides where C lives: *bundled* (shipped inside the artifact, resident on load) or *externalized* (the artifact ships only an index I with one entry e_j per page, $|e_j| = c_j > 0$, $i = \sum_j c_j$, and pages in one routed slice on demand via R9 retrieval over M7 addresses). We prove four claims and one corollary pair. (a) Paged context occupancy is $O(i + s)$ versus bundled $\Theta(\min(B, W))$, with the explicit constant $r = \max_j c_j/\ell(p_j) \leq 1$. (b) Externalization is information-lossless: it is the n-fold composition of M7 evictions, a bijection on stored content, so — whenever the external store answers a dereference (the $a = 1$ idealization; part (d) prices $a < 1$) — the reachable capability set is invariant and the marginal artifact cost of a new capability is one index entry c_{n+1}, not one library $\ell(p_{n+1})$: "adding a capability = adding an index entry" is a theorem of the model, not a slogan. (c) In expected net utility, externalization dominates bundled residency whenever index recall clears an explicit threshold ρ^* — a one-sided sufficient gate, since the comparison deliberately grants bundling full-pool utility; the two limits of that one inequality recover both clauses of the shipped rule — $\rho^* \leq 0$ for any corpus whose residency dwarfs one slice (externalize regardless of index quality), and $\rho^* \geq 1$ when the corpus fits the window and is no larger than its own index plus one average slice (a lean entry doc or 50-line catalog stays inline; externalizing it can never strictly win). (d) Against bundled-but-paged (the library kept in the artifact but read through the same index), every answered-path term cancels, leaving install mass against availability: externalization wins iff $(1-a)\Delta_a < \mu B$, with Δ_a the net failure loss defined in (d).

Setting. Reused objects. From R9: the budgeted-retrieval problem — candidate pool, lengths len, monotone submodular relevance f_t, retrieval budget β (R9's budget, renamed to avoid collision with the library size B), and the cost-benefit greedy with seed enumeration achieving $1 - 1/e$. From M7: the addressed slow tier — index entries are M7 addresses over an alphabet of size D, uniquely decodable (distinct pages receive distinct, unambiguously parseable entries) and byte-gated, $c_j \leq \ell(p_j)$. New variables, disjoint from every other rule's as the coherence extension requires: window W, with $i + s_{\max} \leq W$ (slices are cut to fit — that is what a slice is); task distribution $\mathcal{T}$; routing map σ sending task t to slice $\sigma(t) \subseteq C$ of size $s(t) = \sum_{p \in \sigma(t)} \ell(p) \leq s_{\max}$, mean $\bar{s} = \mathbb{E}_t[s(t)]$; hit event E_t: some budget-β-feasible maximizer of f_t over C lies inside $\sigma(t)$; recall $\rho = \Pr_t[E_t]$; task utility $u_t = g_t(f_t(S_t))$ with g_t non-decreasing in retrieved relevance; residency price $\lambda > 0$ utility per resident reference token (each displaces a task token in the fixed window; linear cost); miss bound $\Delta \geq \mathbb{E}_t\left[u_t^{\text{full}} - u_t^{\text{slice}} \mid \overline{E_t}\right]$, the conditional expected utility gap when routing misses (when $\rho = 1$ the conditioning event is null: the requirement is vacuous, any $\Delta \geq 0$ qualifies, and the miss term below is zero); install price $\mu \geq 0$ per artifact token (the lean-stance cost the benchmark measures; set $\mu = 0$ in part (c), which is therefore conservative, and isolated in part (d)). Access model: retrieval returns a budget-feasible f_t-optimal set from its pool (exact-optimizer idealization; R9's greedy supplies a uniform $1 - 1/e$ implementation on both sides — see Boundary). Net values: $V_{\text{page}} = \mathbb{E}[u^{\text{slice}}] - \lambda(i + \bar{s})$; $V_{\text{bund}} = \mathbb{E}[u^{\text{res}}] - \lambda B_r$ with bundled residency $B_r = \min(B, W)$ and u^{res} the utility of retrieval over the resident portion.

Theorem. Under the Setting: *(a) Space.* For every task, paged occupancy obeys $i + s(t) \leq i + s_{\max} \leq rB + s_{\max}$ with $r = \max_j c_j/\ell(p_j) \leq 1$, versus bundled residency $B_r = \min(B, W)$; hence $(i + s_{\max})/B \leq r + s_{\max}/B \to r$ as $B \to \infty$ at fixed r and $s_{\max}$. *(b) Losslessness.* Externalization is a bijection between the bundled artifact and the pair (index in artifact, pages in external store): no stored information is created or destroyed, every capability reachable bundled is reachable externalized at the cost of exactly one dereference against a store that answers ($a = 1$; (d) prices $a < 1$), artifact occupancy falls from B to i, and capability n+1 costs $c_{n+1} \leq \ell(p_{n+1})$ artifact tokens. *(c) Dominance threshold.* $V_{\text{page}} - V_{\text{bund}} \ \geq\ \lambda\big(B_r - i - \bar{s}\big) - (1 - \rho)\Delta$, so externalization dominates whenever

$$\rho \ \geq\ \rho^* \ :=\ 1 - \frac{\lambda\,(B_r - i - \bar{s})}{\Delta} \qquad (\Delta > 0),$$

and for $\Delta = 0$ whenever $B_r \geq i + \bar{s}$; the condition is sufficient, not necessary — below ρ^* this one-sided bound is silent (see Boundary). Corollaries: (i) if $\lambda(B_r - i - \bar{s}) \geq \Delta$ then $\rho^* \leq 0$ — externalize for every index, however bad; (ii) if $B \leq W$ and $B \leq i + \bar{s}$ then $V_{\text{page}} - V_{\text{bund}} \leq 0$ even at $\rho = 1$ — the lean-entry-doc regime, where bundling weakly dominates even a perfect index. *(d) Install mass.* Against bundled-but-paged (library inside the artifact but read through the same index and selector), the read paths cancel term-by-term and $V_{\text{ext}} - V_{\text{bund-paged}} = \mu B - (1 - a)\Delta_a$, where a is external-store availability and Δ_a is the *net* failure loss — the conditional expected drop in net value when a dereference goes unanswered, utility forgone net of the λ-residency the unfilled slice no longer occupies (a definition, which is what makes the equality exact); externalization dominates iff $(1 - a)\Delta_a < \mu B$, and unconditionally at equal reliability $a = 1$ for any $\mu > 0$ and nonempty corpus ($B > 0$).

Proof. *(a)* The byte gate gives $c_j/\ell(p_j) \le 1$ for each j, so $r \le 1$ and $i = \sum_j c_j \le r \sum_j \ell(p_j) = rB$. Paged occupancy is index plus routed slice: $i + s(t) \le i + s_{\max} \le W$ by the slice-fits assumption, and $i + s_{\max} \le rB + s_{\max}$ by $i \le rB$ (the majorant $rB + s_{\max}$ itself need not fit in W; only the actual occupancy is assumed to). Bundled residency is $\min(B, W)$ by definition of residency in a window of size W. Dividing by B gives the ratio bound; letting $B \to \infty$ at fixed r and $s_{\max}$ sends it to r. No win is claimed yet: for $n = 1$ with the single page routed whole, $i + s(t) = c_1 + \ell(p_1) > \ell(p_1) = B$ since $c_1 > 0$ — part (c) locates the crossover exactly. *(b)* Externalizing page p_j is literally M7's `write=EVICT` with slug e_j: the index is an address book, so by hypothesis the entries form a uniquely decodable code, M7's necessity direction gives $\sum_j D^{-c_j} \le 1$, the sufficiency direction certifies such a code coexists, and each dereference recovers p_j byte-for-byte whenever the store answers (unavailability is priced in (d), not here). By M7, each single eviction is a bijection $(p_j \text{ in artifact}) \leftrightarrow (e_j \text{ in artifact},\ p_j \text{ in store})$ changing artifact occupancy by $-(\ell(p_j) - c_j) \le 0$. The n evictions act on disjoint pages with injective slugs, so their composition is a bijection between the bundled artifact and (index, store): information is relocated, never destroyed; occupancy drops from B to i by $\sum_j(\ell(p_j) - c_j) \ge 0$; and the marginal artifact cost of capability $n{+}1$ is c_{n+1} versus $\ell(p_{n+1})$ — never larger, by the byte gate, and equal to the fraction $c_{n+1}/\ell(p_{n+1})$ of the bundled cost — far below 1 whenever pages dominate their catalog lines. *(c)* First a pointwise fact: $\sigma(t) \subseteq C$, so the budget-feasible subsets of the slice are a subfamily of those of the corpus, the slice optimum of f_t is at most the corpus optimum, and since g_t is non-decreasing, $u_t^{\text{slice}} \le u_t^{\text{full}}$ for every t; likewise the resident portion is a subset of C, so $u_t^{\text{res}} \le u_t^{\text{full}}$ and $V_{\text{bund}} \le \mathbb{E}[u^{\text{full}}] - \lambda B_r$ (when $B > W$ this grant of full-pool utility to bundling is strictly generous, so the comparison is conservative — and one-sided; see Boundary). Now condition on routing. On E_t (probability ρ): some unrestricted budget-feasible maximizer lies inside $\sigma(t)$, so the slice optimum equals the corpus optimum — the exact optimizer over the slice attains the same f_t-value, hence $u_t^{\text{slice}} = u_t^{\text{full}}$. (This is where R9 transfers: its covering-inequality analysis quantifies only over feasible candidates and elements of the optimum, so with the optimum inside the slice the $1 - 1/e$ guarantee holds verbatim on the restricted pool against the *unrestricted* optimum.) On $\overline{E_t}$ (probability $1 - \rho$; if $\rho = 1$ this event is null and the term vanishes): $\mathbb{E}\big[u^{\text{full}} - u^{\text{slice}} \mid \overline{E_t}\big] \le \Delta$ by definition of Δ. Taking total expectations, $\mathbb{E}[u^{\text{full}}] - \mathbb{E}[u^{\text{slice}}] \le (1 - \rho)\Delta$. Therefore $V_{\text{page}} - V_{\text{bund}} \ge \big(\mathbb{E}[u^{\text{slice}}] - \mathbb{E}[u^{\text{full}}]\big) + \lambda(B_r - i - \bar{s}) \ge \lambda(B_r - i - \bar{s}) - (1 - \rho)\Delta$. Non-negativity of the right side rearranges to $(1 - \rho)\Delta \le \lambda(B_r - i - \bar{s})$, i.e. $\rho \ge \rho^* = 1 - \lambda(B_r - i - \bar{s})/\Delta$ for $\Delta > 0$, and to $B_r \ge i + \bar{s}$ for $\Delta = 0$. This is one value-of-information gate: pay the expected miss loss $(1 - \rho)\Delta$ to buy back $\lambda(B_r - i - \bar{s})$ of window. Corollary (i) is immediate: $\lambda(B_r - i - \bar{s}) \ge \Delta$ forces $\rho^* \le 0$. Corollary (ii): for $B \le W$ bundling is fully resident, so $u_t^{\text{res}} = u_t^{\text{full}}$ exactly and $V_{\text{page}} - V_{\text{bund}} = \big(\mathbb{E}[u^{\text{slice}}] - \mathbb{E}[u^{\text{full}}]\big) + \lambda(B - i - \bar{s})$; the first term is ≤ 0 pointwise and the second is ≤ 0 when $B \le i + \bar{s}$, so even a perfect index ($\rho = 1$, first term zero) cannot make externalization strictly better. *(d)* If the library ships bundled but is read through the same index and the same selector, then on the event that the store answers (probability a) pools, hit events, recall, and occupancy $i + s(t)$ are identical term-by-term, so every utility and λ term cancels there; on the failure event (probability $1 - a$) the entire conditional gap between the two variants — utility forgone *net of* the $\lambda s(t)$ residency the unfilled slice does not occupy — equals Δ_a by its definition in the Theorem (without the netting, the residency saved on failure would survive the subtraction and break the equality); and the artifacts differ in install mass by exactly B beyond the shared index. Hence $V_{\text{ext}} - V_{\text{bund-paged}} = \mu B - (1 - a)\Delta_a$ exactly: positive iff $(1 - a)\Delta_a < \mu B$ — the same threshold form as (c) with $(a, \Delta_a, \mu B)$ in place of $(\rho, \Delta, \lambda(B_r - i - \bar{s}))$ — and, at $a = 1$, positive for any $\mu > 0$ since $B = \sum_j \ell(p_j) > 0$ for a nonempty corpus. ☐

In fabius. M9 is the R9/M7 paging gate applied to the system's own packaging: every skill ships a lean entry doc plus index entries (M7 slugs) and pages in one routed slice through R9's density greedy; a `references/` folder that grows past a lean entry doc is tested against $\rho \ge \rho^*$. Bulk corpora — agent catalogs, design teardowns, swipe files, hardening guides, vector stores — carry residency far beyond one slice, so the corollary-(i) premise $\lambda(B_r - i - \bar{s}) \ge \Delta$ is checked by measurement (per the Boundary) and, once it holds, $\rho^* \le 0$: they move out regardless of index quality; a genuine 50-line catalog sits in corollary (ii) and stays inline. Adding a capability adds c_{n+1} tokens of index, not $\ell(p_{n+1})$ of library.

Boundary. PROVED: the occupancy bound (a) with the byte-gate constant $r \le 1$; losslessness (b) as a composition of M7 bijections, inheriting M7's prefix-code machinery wholesale, conditional on the store answering dereferences — availability is not free, it is the a that (d) prices; the dominance threshold ρ^* and both corollaries in (c), under three declared modeling assumptions — linear residency cost λ per resident token, Δ defined as a bound on the conditional miss gap, and the exact-optimizer access model; and the install-mass comparison (d) with its availability threshold. ONE-SIDED: the (c) gate is sufficient, never necessary — bundling is granted full-pool utility and Δ is only an upper bound, so for $\rho < \rho^*$ the theorem is silent (except in the corollary-(ii) regime, where bundling provably weakly dominates); close calls below the gate are decided by measurement, not by this theorem. OPERATIONAL, not proved: every numerical value — ρ must be measured per index as miss-rate over a task log, and $\lambda, \Delta, \Delta_a, \mu, a$ estimated, never assumed; the linearity of λ (real long-context degradation is plausibly convex, which would only widen externalization's win at large B_r, but that is a conjecture, not a theorem here); the exact-optimizer idealization — under R9's greedy the two pools share the same benchmark optimum on hit events and greedy earns the same $1 - 1/e$ guarantee against it on either pool, but pointwise selections may differ inside the band, opening a second-order hit-event gap that Δ (a miss-conditional bound) does not cover: the theorem as proved holds for the exact optimizer, and the greedy implementation needs that gap measured or separately bounded; and the migration inventory itself — which libraries still ship bundled under `references/`, and the count — is a measured fact owned by CORPUS.md, not a consequence of this theorem.

5 · Coherence — the rules are one system

Twenty-two independently derived rules could easily contradict one another. They do not. The capstone result of this paper is that they compose into a single decision system that is **consistent, complete, and composable**.

```
parcus  (lean floor — always on, never competes for a task verb)
  < R1   classify the 3 axes {Memory, Tools, Planning}
  < R4   scout 2–3 routes      (only if the classification is ambiguous)
  < R2   capability ladder      (admit the smallest sufficient rung)
  < R3 · M1   tool / 2nd-agent expected-value gate (confirm or veto the rung)
  < R6   plan-then-bind         (build the tool-free call DAG)
  < R5   reason → act → observe (per-edge execution invariant)
  < R7 · M2   search topology    (branch only if scorable · tree vs graph)
  < R8 · M3 · M4   refine budget  (signal-typed loop · verify depth · retry-stop)
  < M5 · M6   learning layer       (metric-gated rewrite · skill cache, queried first)
  < R9* · M7 · M8*   memory substrate (retrieve-under-budget · evict/recall · rank)
```

Theorem **The policy is one coherent decision system** CAPSTONE

Statement (Coherence Capstone). The eighteen rules $\{\text{R1--R10},\ \text{M1--M8}\}$ compose into a single, well-defined decision system: a function from an incoming task to a terminating sequence of layer dispatches. We prove four properties — consistency, completeness, composability/model-applicability, and isolation of the non-governing rules.

Proof.

(a) Consistency. Strip surface vocabulary and nearly every gate is the *same object*: an expected-loss / value-of-information inequality

$$\mathbb{E}[L \mid \text{don't act}] - \mathbb{E}[L \mid \text{act}] \ > \ \text{cost},$$

instantiated on a *different* capability variable — a tool (R3: $a - b > c_{\text{call}}$), a second agent (M1: $p_a(1 - p_c) > c_{\text{agent}}$), a branch (R7: $\text{EVSI} > c_{\text{search}}$), a refine loop (R8/M4), a corrector depth (M3: $qC_{\text{fail}}r'(0) > \kappa$), a rewrite (M5), a stored fact (M6/M7/R9c/M8c). A family of sign-consistent inequalities, each in its *own* variable over disjoint capability axes, is jointly satisfiable and cannot contradict itself: choosing the truth value of one variable never constrains another. Tie-breaks are uniformly toward the cheaper action (R2's $\leq$, R3/M4's strict $>$, M8c's strict $<$), so no two gates push opposite ways at a tie. The apparent "R2 minimize-machinery vs R7 branch-wide" tension dissolves on variable-counting: R2 sets the *rung-admission bit* (whether a rung enters), R7 sets the *branch-width* (how wide, once admitted) — orthogonal coordinates. Execution invariants R5 (grounding) and R6 (plan-then-bind) constrain the *order* of events, not whether any capability is added, so they live in a third, disjoint coordinate. Memory rules (M6–M8, R9–R9c) act only on the store/window variable. Disjoint variables, one sign convention ⇒ consistent.

(b) Completeness. R1's label space $\{0,1\}^3$ is, by R1's proof, a measurable partition: every task lands in exactly one of 8 cells (mutual exclusivity + exhaustivity of ℓ^{-1}). R2's ladder is a totally ordered chain $0 \leq n \leq N$, and ρ assigns every cell a defined action, including $000 \mapsto$ parcus inline (the $n = 0$ rung). No task shape is unrouted: an empty load still routes (to parcus), and every loaded cell routes to its dominant specialist under $P \succ T \succ M$. The domain is covered with no fall-through.

(c) Composability + model-applicability. Order the firing relation

$$\text{parcus} \prec \text{R1} \prec (\text{R4 if ambiguous}) \prec \text{R2} \prec \text{R3}{\cdot}\text{M1} \prec \text{R6} \prec \text{R5} \prec \text{R7}{\cdot}\text{M2} \prec \text{R8}{\cdot}\text{M3}{\cdot}\text{M4} \prec \text{M5}{\cdot}\text{M6} \prec \text{R9*}{\cdot}\text{M7}{\cdot}\text{M8*}.$$

This is a strict partial order with parcus as floor and no back-edge, hence a finite acyclic DAG; a topological pass visits each rule once, so the policy *terminates*. Each rule's input is produced upstream (R1's bits feed R2; R2's admitted rung feeds R7's width; R6's schedule precedes R5's grounding check), so the composition type-checks. Every rule is stated as a closed-form predicate over quantities an LLM can estimate in-context (loads, costs, gains, hit-rates), so an LLM can execute the entire DAG from the policy text alone — model-applicability.

(d) Isolation of non-governing rules. The one qualitative rule R5 answers only *admissible / inadmissible* (its proof shows no argmax J exists) and is never traded against a score. The operational heuristics — R4 (scout: defer-and-average), R10 (emphasis trades breadth for adherence), M8b (a recency tie-break), and M2's non-associative fallback — carry no objective function and each enters fabius only as a *shape* (defer-and-average, emphasis-trades-breadth, monotone tie-break, merge-needs-a-semigroup). None sits on the load-bearing inequality of (a): they advise a decision the mathematical gates have already made, and removing them changes no gate's truth value. Hence the governing skeleton is exactly the consistent inequality system of (a)–(c). □

In fabius. The capstone certifies that loading fabius installs one decision system, not eighteen competing heuristics: R1 partitions, R2–M8 each test one disjoint cost-benefit variable in a fixed acyclic order with parcus as floor, R5 guards admissibility, and the heuristics advise without ever overriding a measured gate.

Scope, and the completion. The capstone above is stated over the eighteen core rules (R1–R10, M1–M8); with the four §4.7 proofs, the audit below establishes the coherence check over the full *twenty-two-rule* set — threshold form, variable-disjointness, pipeline placement, and contradiction checks per rule. The verdict is **coheres, with exceptions —** and in keeping with the honesty bar, the exceptions are printed in full rather than smoothed away. The frontier layer (R14–R16 · M10–M13) stays outside this theorem by construction.

Extension The theorem over twenty-two rules — R11–R13 and M9 in the pipeline

EXTENSION

The extended firing order. `parcus < R1 < (R4 if ambiguous) < R13 < R2 < R11 < R3·M1 < R6 < R5 < R7·M2 < R8·M3·M4 < M5·M6 < R9*·M7·M8*·M9 — per stage when R13 fires the studio (D < P < E < F), with R11 re-entering dispatch on a verified miss (≤ K−1 escalations per sub-task) and the whole chain wrapped, when work is multi-cycle, by R12's externally capped step→verify loop (≤ N replays).`

Scope. The capstone above certifies the eighteen-rule core as one expected-loss / value-of-information inequality instantiated on pairwise-disjoint variables and composed in a strict acyclic firing order. Four proved entries — R11 (tier cascade), R12 (dual exit gate), R13 (studio composition), M9 (externalization) — complete the system. For each we run the same four checks: (1) threshold form and the variables read and written; (2) variable-disjointness against the existing eighteen, naming every shared object and classifying the reuse; (3) placement in the firing order with acyclicity re-verified; (4) absence of contradiction with any existing gate. Three findings are recorded as exceptions rather than smoothed over: R13 fits the threshold family in form but not in provenance; R11/R12 change the pipeline from a single topological pass to a *bounded iteration* of passes; and R12's hazard coordinate is M4's object at cycle scale, so exactly one pair of gates coheres by proved agreement on an overlap rather than by disjointness. None breaks coherence; each changes the exact wording of the theorem. This result was adversarially re-verified against the four entries, and two slips in the first draft are corrected in place: the worst-case suffix-pass bound (which omitted the per-stage sub-task count) and the misclassification of p_{k+1} as a fresh variable. A second adversarial pass verified every threshold form and tie-break verbatim against its entry and corrected three further slips in place: the overbroad (b′) claim that the extended routing map agrees with the original ρ at every $k = 1$ cell (true only on the legacy $\ell_D = 0$ cells), the description of R11's W as the admitted plan's whole spend (the entry defines it as the expected spend of the plan's *remainder* for the sub-task), and one missed cross-entry interaction — R12's run-level exits can truncate R11's cascade out of its equal-guarantee class, an optimality-scope seam rather than a gate conflict — now recorded as an exception.

R11 — the tier coordinate. *Threshold form.* Attempt tier i ahead of the rest of the plan iff $p_i W > c_i$, equivalently $W > \rho_i := c_i / p_i$. This is R3's gate verbatim under substitution: $a = W$ (expected spend without the probe), $b = (1 - p_i)W$ (residual spend after it), $\text{EVOI} = a - b = p_i W$, $c_{\text{call}} = c_i$. *Variables.* Reads: tier prices c_i, per-sub-task success probabilities $p_i(t)$, downstream spend W — the expected spend of the *remainder* of the admitted plan for the sub-task (the rest of the cascade, or the strong tier directly), an upstream-output reuse of R2's marginal value-versus-cost objects that the entry's placement note declares; an earlier draft of this audit called W the admitted plan's spend wholesale, corrected here — and R8's pass/fail verdict. Writes: the *tier slot* — which model executes the admitted work — a capability axis none of the eighteen occupies. R2 owns the rung-admission bit (how much machinery enters), R3 the tool-call margin (whether an external call is bought), M1 the second-agent bit (whether a peer is spawned), R7 the branch width; R11 decides which brain runs what R2 admitted. Choosing its truth value constrains none of theirs. *Shared objects.* R8's verifier — read-only consumption of the verdict bit as the sole escalation trigger; R11 writes nothing into R8's contraction machinery or thresholds. R3's EVOI functional is reused as threshold *form*, not as a shared variable. *Tie-break.* Strict $>$: at $p_i W = c_i$ the probe is not prepended — the two plans tie in expected spend and the probe-free plan is the leaner action, matching the house convention. *Contradiction check.* R11 vs R2: different questions about the same admitted plan (tier of execution vs level of machinery); no shared coordinate, no possible disagreement. R11 vs R3/M1: the identical inequality on disjoint margins — jointly satisfiable by the capstone's own argument. The route-straight-to-strength clause is not a competing exception rule: it is the same index with ρ_{cheap} inflated through small p or damage-adjusted $\tilde{c}$, so there is exactly one gate, never two pulling opposite ways.

R12 — the loop-exit coordinate. *Threshold form.* Two expected-loss gates: continue iff $(1 - q_k)\, p_{k+1} > c_{\text{cyc}} / W$; at stop, declare done iff $q_k > 1 - c_{\text{esc}} / W$, else escalate. *Variables, classified honestly.* Reads: the dual-gate posterior q_k built from $C_k \wedge D_k$ — genuinely fresh, and the entry's own placement note is exact on this point: q_k is the *only* new variable. The cycle hazard p_{k+1} is **not** fresh: it is M4's hazard object lifted from attempt scale to cycle scale — the continue gate is M4's threshold functional with the belief factor exposed, and this is the single place in the extension where two gates read the same coordinate. The pair cannot disagree: at $q_k = 0$ (a fresh observed failure) the two thresholds coincide exactly — proved in the entry, not assumed — and elsewhere R12 applies the same functional with the same sign, so M4 governs the attempt-level retry after a failure and R12 the run-level cycle decision without any joint assignment being unsatisfiable. But joint satisfiability for this one pair rests on that agreement-at-overlap, not on the disjointness argument; recorded as an exception. Also reads costs $c_{\text{cyc}}, c_{\text{esc}}, W$ and R8's progress metric e_k as the stall test's null (read-only). Writes: the loop-exit decision (continue / stop-done / stop-escalate) — no existing rule owns run-level completion. *Tie-break.* At equality the rule stops — the cheaper action; consistent. *Contradiction check.* R12 vs R8: when R8's contraction holds, R12's stall exit fires falsely with probability at most L^w — the rules agree with quantified error; where R8 offers no guarantee (isometry, martingale), R12 exits, and R8 asserts nothing to the contrary. R12 vs M4's redundancy lemma: verbatim reuse, same sign. No gate anywhere pushes toward continuing while R12's threshold says stop, because no other gate reads q_k at all and the one shared coordinate provably agrees. R12 vs R11 — a seam an earlier draft missed: the stall exit or the hard cap can fire while some sub-task's cascade still holds untried tiers, invoking the fallback early — precisely the move R11's equal-guarantee class forbids and for which R11's entry claims no optimality (its truncation caveat). This is not a gate conflict — both exits route to the same escalation action, so no joint assignment is unsatisfiable — but it scopes R11's cascade-optimality theorem to cycles R12 lets run to completion. The entries' own boundaries meet at the seam: R12's cap analysis ($\mathbb{E}[\sigma^*] \le W/c_{\min}$; the cap is insurance, not the mechanism) certifies the truncated regime is one the optimum already avoids in expectation, and a certified stall is evidence of latent sub-task difficulty — exactly the correlation under which R11's boundary already withdraws the static index's optimality proof. Recorded as an exception.

R13 — the dispatch-topology coordinate. *Threshold form, honestly qualified.* The gate is one threshold on one fresh variable: run the studio iff the loaded-block count $k(t) \ge 2$, where $k(t)$ is computed from R1's load bits extended by a domain bit ℓ_D. In form this implements the house inequality $\mathbb{E}[L \mid \mathsf{collapse}] - \mathbb{E}[L \mid \mathsf{studio}] > c_{\mathrm{pipe}}$, with the loss side supplied by the collapse lemma: the studio output satisfies every published constraint, while a single-layer collapse admits outcomes violating each of the other $k-1$ loaded blocks. But the provenance differs from every other gate in the family, and we state it as an exception rather than force the fit: R13's theorem content is order- and set-theoretic (partition, commuting cylinder intersections, fixed point, goal preservation); the expected-loss gap is *derived* from the structural lemma and is positive only under the mild extra assumptions $h_\mu < 1$, $L_\mu > 0$ on some uncertified block; and the threshold sits at 2 because that is where the collapse lemma first admits a violating outcome (at $k \le 1$ there is nothing to coordinate) — while whether the gap $\sum(1-h_\mu)L_\mu$ actually clears c_{pipe} at that threshold is an in-context estimate, which is why the entry's own Boundary files both $k \ge 2$ and c_{pipe} among its operational items. R13 is a proved composition theorem wearing the threshold uniform, not a calibrated VOI estimate. *Variables.* Reads: R1's bits plus ℓ_D (R1-object reuse, extended, and R1's measurable-partition proof reused verbatim on the ownership map ω). Writes: the dispatch topology (single dispatch vs the ordered studio) — a coordinate no other rule reads: R2 reads the rung, R3 the margin, R7 the width, M6–M8 the store. *The one genuine amendment.* R13 is not purely additive: for $k \ge 2$ it supersedes the original completeness clause's dominant-specialist routing — exclusion under $P \succ T \succ M$ is replaced by ordering under $D \prec P \prec E \prec F$. Dominance survives as *firing order* (who writes before whom reads), not as exclusion (who fires alone); the collapse lemma is precisely the proof that exclusion was the wrong reading for multi-loaded cells. Completeness is re-proved below on the extended label space. *Contradiction check.* Inside each studio stage, R2's ladder, R6 $\prec$ R5's invariants, and every downstream gate fire unchanged, and the locality axiom makes inter-stage contradiction structurally impossible (cylinders over disjoint blocks always intersect nonemptily). The apparent R2-vs-R13 tension ("minimize machinery" vs "run four layers") dissolves by the same variable-counting as R2-vs-R7 in the original capstone: R2 admits rungs *within* a stage, R13 decides how many stages exist — orthogonal coordinates, and at $k \le 1$ R13 itself takes the cheap side (single dispatch, parcus inline at $k = 0$).

M9 — the packaging coordinate. *Threshold form.* Externalize a corpus iff index recall clears $\rho \ge \rho^* = 1 - \lambda\,(B_r - i - \bar{s})/\Delta$ — one value-of-information gate: pay the expected miss loss $(1-\rho)\Delta$ to buy back $\lambda(B_r - i - \bar{s})$ of window; the companion install-mass gate $(1-a)\Delta_a < \mu B$ has the identical form. *Variables.* Reads and writes only fresh coordinates: recall ρ, residency price λ, miss bound Δ, corpus size B, index size i, mean slice $\bar{s}$, install price μ, availability a; the written bit is *where a corpus lives at packaging time* — a time coordinate no run-time rule occupies. M6–M8 and R9* govern what enters the store and window *during* a run; M9 governs the artifact *before* any run. *Shared objects.* R9's budgeted-retrieval machinery (pool, lengths, submodular relevance, budget — renamed β to avoid colliding with B) and M7's uniquely-decodable byte-gated slug code as the index-entry format: both read-only machinery reuse; M9 rewrites no address and evicts nothing at run time — its losslessness claim is literally the n-fold composition of M7's eviction bijections, executed at build time. *Tie-break.* Weak $\ge$: at $\rho = \rho^*$ externalize — toward the leaner resident artifact, the cheap side, consistent with R2's $\le$ convention (and justified by weak dominance: at the tie the bound already gives $V_{\mathrm{page}} \ge V_{\mathrm{bund}}$). *Contradiction check.* M9 vs M8c (strict $<$, store iff cheap): different objects — M8c prices one fact into the run-time store, M9 prices a corpus's residency at build time; no shared variable. M9's corollary (ii), which keeps a lean entry doc inline, agrees in sign with R2/parcus's minimize-machinery bias. One caveat survives as an exception: the gate is *sufficient only* — below ρ^*, outside the corollary-(ii) regime, the theorem is silent and measurement decides.

Shared-object audit. Across the four entries, every shared object is read-only or form-level reuse except one shared coordinate, and none is a write conflict:

- R8's verifier verdict is now read by three rules — R8's own refine loop, R11's escalation trigger, R12's gate signal C_k and stall-test null. No rule writes into another's machinery, so disjointness holds; but the verifier's approximate soundness becomes a *shared single point of failure*, recorded as an exception: one false pass simultaneously erodes R11's equal-guarantee class, R12's run-level false-stop bound, and R8's own contraction certificate, and the disjoint-variables argument certifies nothing about this correlated mode.
- M4's hazard functional is reused by R12 at cycle scale with the belief factor exposed — the one *shared coordinate* in the extension. It coheres by the entry's proved agreement at the overlap (exact coincidence at $q_k = 0$, same functional and sign elsewhere), not by disjointness; recorded as an exception because (a′)'s blanket disjointness wording must carry this qualification.
- R1's load bits and partition proof are reused by R13, extended by one bit — reuse of a proof and an upstream output, not a second writer to the same variable.
- R2's marginal value-versus-cost objects feed R11's W (the expected spend of the admitted plan's remainder for the sub-task), and R3's EVOI threshold is reused by R11 as functional form — input and form reuse; R11's decision coordinate (the tier slot) is touched by neither.
- R9's retrieval objects and M7's slug code are reused by M9 as mechanism and format, with M9's own decision variable untouched by either.
- Symbol collisions are not variable collisions: ρ_i (R11's cost-per-success index) vs ρ (M9's recall); W (R11's downstream spend) vs W (R12's false-stop loss); Δ (M9's miss gap) vs Δ_a (M9's availability loss); L (R8's contraction modulus) vs L_μ (R13's miss cost); β_C, β_D (R12's miss rates) vs β (R9's retrieval budget as M9 renames it); q (R11's slot-reach probability in the exchange lemma) vs q_k (R12's completion belief). All denote distinct variables on distinct axes; the whitepaper's notation table should disambiguate.

Pipeline and acyclicity. Per pass, the firing relation extends to

$$\mathrm{parcus} \prec \mathrm{R1} \prec (\mathrm{R4\ if\ ambiguous}) \prec \mathrm{R13} \prec \mathrm{R2} \prec \mathrm{R11} \prec \mathrm{R3{\cdot}M1} \prec \mathrm{R6} \prec \mathrm{R5} \prec \mathrm{R7{\cdot}M2} \prec \mathrm{R8{\cdot}M3{\cdot}M4} \prec \mathrm{M5{\cdot}M6} \prec \mathrm{R9^{*}{\cdot}M7{\cdot}M8^{*}}\cdot$$

Three placements are forced by type-checking, one by convention. R13 must follow R4, not just R1 — R4 finalizes ambiguous labels and R13 counts loaded blocks from finalized labels — a placement the R13 entry leaves implicit, which we fix here and which R11's own placement string ($\mathrm{R1} \prec \mathrm{R4?} \prec \mathrm{R2}$) corroborates. R11 sits between R2 and R3·M1: it consumes R2's admitted plan (its W is the expected spend of that plan's remainder for the sub-task) and prices tiers before any call fires. M9 joins the terminal packaging stage, consuming R9/M7 machinery produced upstream. The conventional one is the studio's stage order itself: when R13 fires, the entire downstream suffix runs once per stage in $D \prec P \prec E \prec F$, an order R13's own boundary (a) marks as operational rather than set-theoretic (the T_λ commute); ordering pairs (stage, rule) lexicographically keeps the composition a finite strict partial order. Two edges re-enter earlier stages, and this is the substantive change to clause (c): R11's verified-miss escalation returns to dispatch (attempt counter $j \le K$, per sub-task), and R12 wraps the whole chain (cycle counter $k \le N$, externally capped, checkpointed). Neither is a back-edge *within* a pass — R12's C_k consumes R8's verdict from the just-completed cycle, R11's escalation consumes the just-completed verification — so each re-entry strictly increments a bounded counter. Unrolling on $(k, \mathrm{stage}, \mathrm{sub\text{-}task}, j, \mathrm{rule})$ in lexicographic order exhibits the full firing relation as a finite DAG with at most $N \cdot 4 \cdot m \cdot K$ passes of the finite suffix, where m bounds the sub-tasks a stage dispatches — R11's cascade runs per sub-task, a factor a first draft of this audit omitted. Acyclicity and termination are preserved, but the original clause "a topological pass visits each rule once" must be restated as "once per pass, boundedly many passes."

Statement (Extended Coherence Capstone). The twenty-two rules $\{\mathrm{R1\text{--}R13},\ \mathrm{M1\text{--}M9}\}$ compose into a single, well-defined decision system: a function from an incoming task to a terminating sequence of layer dispatches, organized as a boundedly iterated acyclic pipeline. We prove the four capstone properties in extended form.

(a′) Consistency. Counting per rule, twenty-one of the twenty-two gate on instances of the one inequality $\mathbb{E}[L \mid \text{don't act}] - \mathbb{E}[L \mid \text{act}] > \text{cost}$ (R12 and M9 each contribute two thresholds of the identical form): the eighteen original coordinates plus the tier slot (R11: $p_i W > c_i$), the loop-exit pair (R12: $(1-q_k)p_{k+1} > c_{\text{cyc}}/W$ and $q_k > 1 - c_{\text{esc}}/W$), and the packaging bit (M9: $\rho \ge \rho^*$). R13 contributes a gate of the same threshold form on a fresh variable ($k(t) \ge 2$) whose loss side is supplied structurally by the collapse lemma — the one member of the family whose inequality is derived rather than estimated. All shared objects are read-only or form-level reuse except the single shared coordinate p_{k+1} (M4/R12), and that pair coheres by the entry's proved agreement at the overlap rather than by disjointness. Every tie resolves toward the cheaper action (R11 strict $>$, R12 stop-at-tie, R13 single-dispatch at $k \leq 1$, M9's $\geq$ toward the leaner artifact). A family of sign-consistent inequalities on disjoint variables is jointly satisfiable, and the one non-disjoint pair is jointly satisfiable by agreement: no gate's truth value constrains another's. One further cross-entry seam — R12's exits truncating R11's cascade — affects optimality scope only, never joint satisfiability: both gates route the truncated case to the same escalation action, so consistency is untouched (see the exception).

(b′) Completeness. R1's label space extends to $\{0,1\}^3 \times \{0,1\}$ via the domain bit; by R1's partition proof, lifted verbatim by R13, the sixteen cells remain a measurable partition. The routing map ρ' is total: $k=0$ routes to parcus inline, $k=1$ to the unique loaded owner — on the eight legacy $\ell_D = 0$ cells this coincides with the original ρ; the eight new $\ell_D = 1$ cells (including the domain-only cell at $k=1$, which routes to the domain leader) are routes the original map never defined, introduced with the domain bit, not re-routings of anything it did define — and $k \geq 2$ to the studio order with unloaded stages firing as no-ops. Dominance survives as firing order rather than exclusion. No task shape is unrouted; no fall-through exists.

(c′) Composability + model-applicability. The displayed per-pass relation is a strict partial order with parcus as floor and no back-edge within a pass. R11 and R12 introduce re-entry along strictly increasing, bounded counters ($j \leq K$ tiers per sub-task, bounded structurally by the finite ladder; $k \leq N$ cycles enforced by the harness, not the model), so the unrolled firing relation is a finite DAG and the policy terminates — in at most $N \cdot 4 \cdot m \cdot K$ suffix passes in the worst case, with R12's own bound $\mathbb{E}[\sigma^*] \leq W/c_{\min}$ certifying the cap's scale. Every input is produced upstream or in a prior pass: R4's finalized bits feed R13's count, R13's stage list feeds R2, R2's admitted plan feeds R11's W (as the plan-remainder spend per sub-task), R8's verdict feeds R11's escalation and R12's gate, R9/M7's machinery feeds M9. Every gate is a closed-form predicate over in-context-estimable quantities, so an LLM executes the full system from the policy text alone.

(d′) Isolation. Unchanged, and strengthened by absence: none of the four new entries is a heuristic — all four are load-bearing proved gates, so the governing skeleton grows from eighteen coordinates to twenty-two while the advisory set (R4, R10, M8b, M2's fallback) stays fixed and still advises decisions the measured gates have already made. R5 remains the sole qualitative admissibility guard, never traded against a score, and its position is untouched. ☐

In fabius. Loading fabius installs one decision system, not twenty-two competing heuristics: R1+R13 partition and shape the dispatch, R2–M9 each test one cost-benefit variable in a fixed per-pass order with parcus as floor, R11 prices which brain runs each admitted sub-task, R12 bounds how long the whole pipeline may cycle — with the harness, not the model, holding the cap — and M9 prices where the system's own reference mass lives. The exceptions below are the exact places where the extended theorem's wording differs from the original, and they are wording, not fracture: every one is a qualification the entries themselves prove or declare, not a conflict between gates.

Stated exceptions, in full. The verdict is *coheres, with exceptions* — and the exceptions are part of the theorem, not a footnote:

- R13 fits the threshold family in form only: its gate k(t) ≥ 2 is a genuine threshold on a fresh variable, but the rule is structural — the expected-loss gap is derived from the set-theoretic collapse lemma and is positive only under the extra assumptions h_mu < 1 and L_mu > 0 on some uncertified block. Structure locates the threshold (k ≥ 2 is exactly where the collapse lemma first admits a violating outcome; at k ≤ 1 there is nothing to coordinate), but whether the gap Σ(1−h_mu)L_mu clears c_pipe at that threshold is an in-context estimate — the entry's own Boundary files both k ≥ 2 and c_pipe among its operational items. R13 is a proved composition theorem wearing the threshold uniform, not a calibrated VOI gate.
- R13 amends rather than disjointly extends the original completeness clause (b): for multi-loaded cells the dominant-specialist exclusive routing (P > T > M) is superseded by the studio order — dominance survives as firing order, not exclusion — so the routing map ρ is retired for k ≥ 2 and completeness had to be re-proved on the extended {0,1}^4 label space (done in (b′)). One qualification this audit adds: the extended map ρ' coincides with the original ρ only on the eight legacy ℓ_D = 0 cells; the eight ℓ_D = 1 cells (including the domain-only k = 1 cell) are new routes introduced with the domain bit, not re-routings of anything the original map defined.
- The capstone clause 'a topological pass visits each rule once' weakens to 'once per pass, boundedly many passes': R11 (≤ K−1 verified-miss escalations per sub-task) and R12 (≤ N externally capped cycles) make the composed system a bounded iteration of the acyclic pipeline. Termination is preserved — the counters strictly increase, the caps are enforced structurally (finite tier ladder, finite sub-task list) or by the harness (N), and the

firing relation unrolled on (cycle, stage, sub-task, attempt, rule) is a finite DAG — but the single-pass wording of (c) is no longer literally true, and the worst-case suffix-pass count is $N \cdot 4 \cdot m \cdot K$ with m the per-stage sub-task count (R11's cascade runs per sub-task), not $N \cdot 4 \cdot K$ as a first draft of this audit stated.

- R8's verifier verdict is now read by three rules (R8's own loop, R11's escalation trigger, R12's gate C_k and stall-test null). This is read-only reuse, not a write conflict, but it makes the verifier's approximate soundness a shared single point of failure: one false pass simultaneously erodes R11's equal-guarantee class, R12's run-level false-stop bound, and R8's contraction certificate. The disjoint-variables argument certifies nothing about this correlated failure mode; R12's dual gate mitigates but does not eliminate it.
- R12's hazard coordinate p_{k+1} is not a fresh variable — R12's own placement note declares q_k the only new variable, and p_{k+1} is M4's hazard object lifted from attempt scale to cycle scale: the single shared coordinate in the extension. Joint satisfiability for the pair (M4, R12) therefore rests not on disjointness but on the entry's proved agreement at the overlap — at $q_k = 0$ (a fresh observed failure) the two thresholds coincide exactly, and elsewhere R12 applies the same functional with the same sign, so the pair cannot disagree. Clause (a′) carries this one agreement-on-overlap qualification; an earlier draft of this audit called p_{k+1} 'fresh', corrected here.
- R12's run-level exits interact with R11's optimality scope: the stall exit or the hard cap can fire while a sub-task's cascade still holds untried tiers, invoking the fallback early — exactly the move R11's equal-guarantee class forbids and for which R11's entry explicitly claims no optimality (its truncation caveat). No gate conflict results — both exits route to the same escalation action (R7/M2), so consistency and joint satisfiability are untouched — but R11's cascade-optimality theorem is thereby scoped to cycles R12 lets run to completion. The mitigation is internal to the entries: R12's cap-scale bound ($E[\sigma^*] \leq W/c_{min}$; 'the cap is insurance, not the mechanism') certifies the truncated regime is one the optimum already avoids in expectation, and a certified stall is evidence of latent sub-task difficulty — the correlation under which R11's own boundary already withdraws the static index's optimality proof. The two entries' caveats meet at this seam rather than contradict. An earlier draft of this audit missed the interaction; added here.
- M9's gate is one-sided sufficient, never necessary: below ρ^* the theorem is silent (except in the corollary-(ii) lean regime, where bundling weakly dominates even a perfect index), so packaging decisions in the band $\rho < \rho^*$ are settled by measurement, outside the coherence system.
- Symbol collisions across entries are not variable collisions but require notational care in the whitepaper: ρ_i (R11's cost-per-success index) vs ρ and ρ^* (M9's recall and threshold); W (R11's downstream spend) vs W (R12's false-stop loss); Δ (M9's miss gap) vs Δ_a (M9's availability loss); L (R8's contraction modulus) vs L_{mu} (R13's block miss cost); β_C, β_D (R12's miss rates) vs β (R9's retrieval budget as M9 renames it); q (R11's slot-reach probability in the exchange lemma) vs q_k (R12's completion belief). All are distinct variables on distinct axes.
- The extended theorem certifies consistency, completeness, and termination of the composition — not unconditional optimality of each gate. R11's cascade optimality is conditional on independent tier outcomes (A1) and calibrated (c_i, p_i); R12's multiplicative false-stop drop is conditional on separately-sourced (conditionally independent) C and D, its first-crossing optimality on the monotone-hazard hypothesis and the declare-done loss; these conditions are inherited from the entries and bound what coherence means here.
- One placement was fixed by this audit rather than by the entries: R13's stated placement (R1 < R13 < R2) omits R4; type-checking forces R4 (when ambiguous) to fire before R13, since R13 counts loaded blocks from finalized labels — and R11's own placement string (R1 < R4? < R2) corroborates the insertion. The pipeline above adopts R1 < (R4) < R13 < R2 accordingly.

6 · Does it work? The fabius benchmark

Fabius is prompt-level scaffolding, so a number measured on one model at one moment does not transfer cleanly. The defensible thing to publish is the **method, the mechanism, and the dated measured signal —** plus a one-command way to re-measure it. There is one benchmark, read out on four panels: blind model-judged quality on the four Claude tiers used in the 2026-07-01 run; a mixed historical model-operated execution-score and model-graded checklist track; aggregate-only blind external-model demos; and the **FBS run**, a versioned 100-task suite scored by model judges and a model factual-check grader. Panel A improved three of four measured Claude tiers and reduced output on all four. Panel B's reported executed-code score rows tied at ceiling; its measured lift comes from model-graded checklist scores. The old receipt lacks candidate code and stdout/stderr, so those execution scores are not independently replayable. Panel D prints gains and regressions. No panel establishes an every-model guarantee.

6.1 · The method

One test, three arms. Each task is answered three ways: `baseline` (the task only), `terse` (the task plus a generic "*be concise, write minimal code*" line), and `fabius` (the task plus the shipped stance — the shipped `AGENTS.md` plus the routed `SKILL.md`, read verbatim). The `terse` arm is the real control: beating `baseline` is easy; **beating `terse` isolates structure from mere brevity.**

Three of the benchmark's four panels read out on this three-arm design. The **quality panel** is blind model-judged: **two** judge models that are never told which arm produced which answer score each answer 0–5 on correctness, minimality, and best-practice, plus a deterministic character count; the judges disagree by 0.72/15 on average. Panel B is mixed: its historical code track was extracted, run, and reported by a tool-using model operator, while its domain deliverables were interpreted against fixed factual checklists by two model graders. The current harness uses a deterministic local code runner and retains answer hashes, code, stdout/stderr, and votes for future runs; it cannot repair the old receipt. The **external panel** carries the same three arms cross-family through a portable harness, but only its aggregate is committed. The **FBS panel** swaps the arms for `BASE` / `FAB` / `FAB_MEMORY` and is scored /28 by two blind model judges plus a model factual-check grader. A deterministic structural suite derives its own invariant count and checks the recursive install surface, single-owner contracts, references, and content-bound seal. The canonical aggregate is `evals/results.benchmark.json`; v5/v6/v7 retain scores, not the original answers, and Panel C has no committed raw receipt.

6.2 · Panel A — quality, blind

Panel A — 15 tasks × 3 arms on the four Claude models current at the 2026-07-01 run (Fable 5 · Opus 4.8 · Sonnet 5 · Haiku 4.5); the `fabius` arm is the shipped `AGENTS.md` plus the routed `SKILL.md`, read verbatim; **two** blind judges (inter-judge gap 0.72/15), /15. Raw receipt: `evals/results.v5.json`.

Model	baseline	fabius	Δ quality	output cut
Fable	14.50	**14.73**	+0.23	−25.3%
Sonnet 5	14.07	**14.50**	+0.43	−33.7%
Opus	14.40	**14.60**	+0.20	−20.0%
Haiku	11.73	11.40	−0.33	−35.5%

Output fell 20–35.5% on each of these four measured tiers. Fable, Opus, and Sonnet 5 beat *both* controls; on Sonnet 5 the lift (**+0.43**) is the largest of the four tiers. The one miss stays on the record: Haiku dips overall. On its twelve specialist tasks Haiku gains +0.71 on average, while its three trivial one-liners average −4.50. The pooled category receipt also contains regressions in build, accessibility, and game tasks. These observations motivate model-tier routing and a leaner route; they do not prove those mechanisms fix the regressions. The historical receipt preserves every score cell but not the answer text, so it cannot be independently re-judged.

6.3 · Panel B — mixed verification

Panel B — four historical code tasks executed and reported by a tool-using model operator, plus five domain deliverables interpreted against fixed factual checklists by two model graders; 9 deliverables × 3 arms on the same four models. The score receipt omits candidate files and raw stdout/stderr. Receipt: `evals/results.v6.json`.

Model	baseline %	fabius %	Δ passed
Haiku	75.6	**93.0**	+17.4
Opus	84.9	**90.7**	+5.8
Sonnet 5	84.9	**90.7**	+5.8
Fable	87.2	**90.7**	+3.5

The retained code score rows tie at 100% for baseline and fabius on all four models, so that track contributes no measured lift. The model-graded domain checklist scores rise: Haiku 58→88, Sonnet 5 74→84, Opus 74→84, Fable 78→84. Per deliverable, the SQL checklist moves 67.5%→100%, webhook 40%→67.5%, and Solana account validation 47.5%→57.5%. The combined mixed score rises +3.5 to +17.4 while output falls 12–25%, entirely because the model graders marked more checklist items present. This is directional model-mediated evidence, not a deterministic correctness proof; the historical code artifacts and stdout/stderr are unavailable for replay.

6.4 · Panel C — external-model demos, and what the data supports

Panel C — external-model demos: the portable harness (`evals/portable_eval.py`) carries the same three arms cross-family, blind; 6 tasks, measured 2026-06-22 with live provider keys. The signal is the lift vs the "*be concise*" control on genuine-build, /15.

Family	lift vs terse (genuine-build, /15)
Grok	**+8.5**
GPT	**+7.0**
Claude	**+7.0**
Mistral	**+2.5**

Each of the four measured families has a positive aggregate in this dated demo. The raw portable receipt was not committed, so the published cells cannot be independently replayed; a rerun on moving endpoints is a new measurement. Gemini is wired into the harness, and no Gemini number is printed without a measured run.

6.5 · Panel D — the FBS run: BASE / FAB / FAB_MEMORY on a versioned suite

The system now carries its own identity-and-evaluation contract, `IDENTITY.md`: fabius is an *intelligence amplification layer*, not a model, so the benchmark question is fixed as "*does the exact same model achieve better outcomes with less waste?*" — and the executable form of that contract is the **Fabius Benchmark Suite** (`evals/suite/`, FBS v1.0). One hundred neutral, production-shaped tasks — 20 smoke, 50 core, 30 stress across ten categories — each carries 3–6 fixed textual `automatic_checks` and, where memory matters, a committed `memory_snapshot`. Each task runs as `BASE`, `FAB`, and `FAB_MEMORY`. Two blind model judges score the seven-dimension rubric /28; another model grader interprets the fixed checks from answer text. Character count is deterministic. The historical receipt preserves score rows, not candidate answers or individual judge votes, so its aggregates replay but its judgments cannot be independently repeated.

Panel D — the FBS run, 2026-07-05. Sonnet 5 on the full 100-task suite; Haiku 4.5 on the 20-task smoke tier. Model-judged rubric /28, model-graded factual checks, deterministic mean output chars. Score receipt: `evals/results.v7.json`.

Model · mode	rubric /28	checks	output	vs BASE
Sonnet 5 · BASE	26.14	93.2%	4,001	—

Model · mode	rubric /28	checks	output	vs BASE
Sonnet 5 · FAB	**26.16**	**93.5%**	**3,580**	**held · −10.5% output**
Sonnet 5 · FAB_MEMORY	**26.16**	93.3%	**3,512**	**held · −12.2% output**
Haiku 4.5 · BASE	25.27	96.1%	1,531	—
Haiku 4.5 · FAB	**26.00**	**97.1%**	**1,315**	**+0.73 · −14.1% output**
Haiku 4.5 · FAB_MEMORY	**26.73**	96.1%	**1,361**	**+1.46 · −11.1% output**

Four results. **(i)** On the fast tier the model-judged rubric rises +0.73 from the stance and +0.73 more from memory, on 11–14% less output. **(ii)** On Sonnet 5 the rubric is nearly flat while output falls 10–12%; on stress tasks the model-graded factual-check rate rises 89.9%→93.7%. **(iii)** The core tier is the printed miss: `FAB_MEMORY` −0.60, and `FAB` −0.48, driven largely by one 363-character stub scored 1.5/28 against 28/28 for the other modes. **(iv)** Memory cuts both ways: the model-judged design/product category rises +4.91, while security (−1.57) and error recovery (−1.37) fall. These are dated model-graded measurements with explicit regressions, not an objective oracle.

The honest claim: **in this dated benchmark, Panel A improved three of four measured Claude tiers while shortening all four; the other panels show a mixture of model-graded gains, ties, and regressions.** Not "smarter." Not "10× on everything." The caveats are load-bearing: samples are small; all quality, checklist, and FBS scores are model-mediated; the historical receipts omit candidate answers, Panel B execution artifacts, and Panel C raw data; and every roster is dated. Character counts and future local code execution are deterministic, but they do not retroactively upgrade the historical evidence.

7 · The honesty ledger

The quality bar of the whole project is one phrase: **measured, not claimed.** It runs in two places. In the mathematics, a rule earns a place in the theorem only with a proof that survived adversarial checking; the few operational heuristics (R4, R5, R10, M8b) are labelled as heuristics, carry no proof claim, and never govern a route the way the mathematical gates do. In the benchmark, every number is reported with its method, its controls, and its misses — the losses printed in the receipt, not smoothed away.

Most of the twenty-two rules reduce to one value-of-information threshold that genuinely governs the decision; the remaining few are operational heuristics that advise without ever overriding a measured gate. Every rule is fabius's own — derived, proved where a proof is claimed, and stated as a heuristic where it is not.

The same discipline draws the line between the *proven core* and the *working edge*: a rule belongs to the theorem only with a proof that survived adversarial checking — and the composition's verdict ships with its exceptions printed in full. The researched frontier layer (R14–R16 · M10–M13) sits at that edge — sound, honesty-tagged, and outside the proof until its own debt is paid. The boundary is the honest thing, and it is drawn in ink, not pencil.

8 · Reproduce it, and use it

Every figure in §4 is computed from committed source — `numpy` to SVG. The committed v5/v6/v7 score rows replay into the published aggregates through `node evals/verify-receipts.mjs`; they do not recreate missing historical answers or Panel B execution artifacts. Panel C's aggregate-only result cannot be replayed. New harness runs produce new, dated measurements.

```
# the figures
python3 assets/charts/render_figures.py

# the benchmark — wiring check (no key, no cost), then a real run
node evals/eval.mjs --selftest
ANTHROPIC_API_KEY=...  node evals/eval.mjs --model claude-sonnet-5
GEMINI_API_KEY=...     python evals/portable_eval.py --models gemini
```

Fabius is a plugin with a committed Claude marketplace manifest, fifteen plain-Markdown skill contracts, and an `AGENTS.md` bridge for tools that read standing instructions. The recursive structural gate checks that compatible discovery sees exactly those fifteen contracts. Harness-specific installation and schema support still need to be verified against the harness version being used; the repository-local runtime is an optional component of a full checkout, not a standalone npm package.

The system is built to grow without bloating. Adding a capability is a new `skills/fabius-<name>/SKILL.md` with one precisely-described owned concern plus a row in the corpus index — never a megabyte bundled into the install (M9). That is how the system went from six skills to fifteen — adding the go-to-market, defensive-security, game-craft, on-chain, automation, science, AI/ML-engineering, markets, and cross-model-council verticals — while the installed brain stayed lean and the libraries moved behind the index. Each new layer links to its siblings instead of duplicating them, so the single-owner contract, and the coherence it enables, survive every addition.

9 · Conclusion

Fabius is a small idea carried all the way down. The idea is a single axis — scout wide, strike narrow — that lets an agent be thorough and minimal at once because the two live on different axes. The system is fifteen coordinated skills with a single-owner contract. The policy is twenty-two proven rules — each derived from fabius's own decision-theoretic core — and a researched frontier layer held honestly at the edge. And the mathematics is not decoration: every gate is the same value-of-information threshold applied to a different capability, composed in an acyclic pipeline that routes every task and terminates — proved here, and adversarially checked.

What makes it trustworthy is the line it refuses to cross: a rule earns the theorem only with a proof that survives adversarial checking, and the few operational heuristics are labelled as such. The benchmark applies the same restraint: three of four measured Panel A tiers improve while all four shorten; Panel B's historical code scores tie and its model-graded checklist scores rise; Panel C lacks raw receipts; Panel D prints gains and regressions. **Measured, with the limits visible.**

Scout wide, strike narrow.